

Center for Combined Smart Energy Systems

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Building Energy Management System Based on Multi-Agent Deep Reinforcement Learning

Master's Thesis submitted for the degree of Master of Science
(M.Sc.) in Energy and Process Engineering

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Abstract

With the increasing deployment of residential photovoltaic (PV) systems, battery energy storage systems, and electric vehicles (EVs), prosumers in low-voltage distribution networks are gaining both the need and the capability to manage their electricity consumption and generation flexibly. The energy scheduling decisions of multiple prosumers affect not only their individual operating costs but also the safety of the distribution network. Coordinated control under fluctuating electricity prices, EV mobility requirements, and grid safety constraints therefore represents an important challenge in building energy management.

This thesis investigates the coordinated scheduling of EV charging, stationary batteries, and PV generation for three prosumers connected to a low-voltage distribution network. A distribution network simulation environment is developed using pandapower and incorporates physical models of stationary batteries, EVs, PV systems, and distribution network safety constraints. Multi-agent deep reinforcement learning (MADRL) is employed to learn price-aware and resource-coupled control policies. Action feasibility checks, EV departure constraint handling, and an optional grid safety projection are applied before the actions are executed. The MADRL variants are compared with Local model predictive control (Local MPC), alternating direction method of multipliers-based MPC (ADMM-MPC), and a rule-based battery controller.

The controllers are evaluated over 15 days for three prosumers, resulting in 45 EV departure records. Among the investigated MADRL variants, Projected Hard provides the best overall balance between economic performance, EV feasibility, and distribution network safety. Its total cost is 186.04 EUR, which is 32.1% lower than the 274.01 EUR achieved by the rule-based controller. However, it remains 188.90 EUR and 222.41 EUR more expensive than ADMM-MPC Hard and Local MPC Hard, respectively. Projected Hard reduces the number of voltage violations from three for the rule-based controller and 25 for Local MPC Hard to zero; ADMM-MPC Hard also records zero voltage violations, so that it can ensure grid safety. Moreover, Projected Hard satisfies the EV departure state-of-charge target in all 45 departure records, which behaves better compared with MPC methods.

The results show that MADRL can learn more flexible price-aware and resource-coupling

policies than the rule-based controller. A progress penalty improves EV feasibility under soft constraints by replacing a sparse departure signal with denser temporal guidance, whereas hard constraint handling enforces EV charging feasibility at the action-execution layer. Grid safety projection eliminates voltage violations in the evaluated scenario, but modifies the actions selected by the learned policy and consequently reduces battery arbitrage performance. Overall, the results demonstrate the trade-off among economic cost, EV mobility requirements, and distribution network safety, and provide an experimental basis for coordinated EV–battery–PV control in low-voltage networks with multiple prosumers.

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1 Introduction

With the continuous growth of photovoltaic (PV) systems, battery energy storage systems (BESS), and electric vehicles (EVs) in residential applications, users in traditional distribution networks are gradually evolving from passive electricity consumers into prosumers capable of electricity generation, energy storage, and flexible electricity consumption. For an individual household, PV generation can reduce electricity purchased from the grid, the stationary battery can be charged during low-price periods and discharged during high-price periods, and EV charging can also be scheduled according to user mobility requirements and electricity price signals. However, when more prosumers are connected to the same low-voltage distribution network, these flexible resources can no longer be treated only as an economic optimization problem within individual prosumers. The EV charging, battery charging and discharging, and PV utilization of multiple households can modify the total net load, affect bus voltages, line loading, and transformer loading. Therefore, building energy management should consider economic performance from the user side, EV mobility requirements of the prosumers, and total network safety.

This thesis investigates a multi-prosumer energy management problem in a low-voltage distribution network. The system consists of 3 prosumers, each equipped with its own household loads, PV generation, a stationary battery, and EV charging demand. A simulation environment based on pandapower is developed to map the net load of the prosumers to distribution network power flow calculations, thereby evaluating distribution network safety indicators such as bus voltages, line loading, and transformer loading. The control framework in this work integrates LSTM forecasting, a MADRL actor-critic controller, EV soft or hard constraint handling, and grid safety projection, and compares their performance with baseline controllers. The main objective is to analyze how different control methods can influence the total cost, battery arbitrage, EV charging cost, departure SoC feasibility, and distribution network safety.

This thesis is organized as follows: Chapter 2 reviews the literature on building energy management and different controlling methods used today, and identifies the research gaps. Chapter 3 shows the system model of the thesis, including the models of prosumer loads, PV generation, stationary batteries, EVs, electricity prices, net load, power flow, and the optimization objective. Chapter 4 introduces the methodology, including LSTM

forecasting for electricity price, the MADRL observation and action design, the reward function of MADRL, safety projection, different EV constraint handling methods, and the baseline controllers. Chapter 5 describes the experimental datasets, training settings, network configuration, and evaluation metrics for the experiments. Chapter 6 presents and analyzes the experimental results of the different controllers. Finally, Chapter 7 summarizes the main findings of this thesis and discusses outlook for future research.

2 State of the Art

This chapter presents the research topic of this thesis, including the building energy management in distributed networks, the existing control methods, and the remaining research gaps. The focus of this thesis is the coordinated energy management of multiple prosumers in a low-voltage distribution network. Each prosumer consists of a household load, PV generation, a BESS, and residential EV charging demand. The controller should balance the dynamic electricity price, renewable energy fluctuations, EV departure SoC requirements, and grid safety constraints. Therefore, this chapter first introduces the fundamental of HEMS, then reviews the existing control methods, and finally summarizes the limitations and challenges of these methods, thereby motivating the research presented in this thesis.

2.1 Building Energy Management in Distributed Networks

With the development of smart grids, advanced infrastructure and more residential distributed energy resources, household users have gradually evolved from traditional passive loads into prosumers that can more actively participate in distribution network operation. Smart HEMS is a key implementation of residential demand-side management. It can monitor and coordinate household loads, PV systems, BESS, EVs, and other energy resources in the real time, thereby reducing electricity costs and improving energy efficiency. Nowadays, HEMS has gradually evolved from simple load scheduling under the traditional centralized power supply framework toward cyber-physical HEMS with communication, sensing, control, and user response capabilities [34]. Besides, the widespread deployment of residential air conditioners and other devices has made HEMS become a dynamic multi-objective optimization problem, so that it can simultaneously balance electricity cost, thermal comfort and other objectives. Since users' demand of "optimal" energy usage is very subjective and time-varying, HEMS need to adjust its control objectives according to user preferences [9].

In building energy management, PV systems, BESS, and EVs are among the most important flexible energy resources, therefore we mainly focus on these three devices in this

thesis. PV generation can reduce the electricity demand purchased from the distribution network. When residential PV systems are jointly managed with household loads and other energy resources, their economic value can be improved under market mechanisms. But they are inevitably affected by the intermittency of PV generation and the uncertainty of load demand [6]. Besides due to its fluctuation, it can also cause reverse power flow and voltage problems during periods of low demand and high generation. As for batteries, Dongol et al. investigated MPC-based peak shaving using residential batteries in a rural low-voltage distribution network, showing that residential batteries can be used to smooth PV feed-in and load peaks, although their effectiveness depends on the availability of prediction information and the design of the control objectives [8]. The integration of BESS also enables other applications. Manjunatha et al. demonstrated the application potential of battery energy storage systems in distributed distribution networks for peak shaving, cost-optimal control, and operation under time-varying electricity prices [16]. Id Omar et al. further discussed stacked control of behind-the-meter BESS for both local services and distribution network services, including PV self-consumption, peak-load reduction, and balancing services, indicating that BESS has evolved beyond a household cost optimization tool and has become a flexible resource for providing grid services [10].

At present, except for BESS and PV, EV is also more and more important. Residential charging stations have enabled EVs to become part of HEMS, while the large-scale integration of EVs has further increased the complexity of energy management. Similar to BESSs and PV systems, EV can also be regarded as flexible resources, and their charging schedules can be adjusted according to electricity prices and user mobility requirements. Unlike BESSs and PV systems, however, EVs are characterized by the connection periods, charging power limits, and departure SoC requirements. Fast EV charging stations may also have an impact on residential distribution networks. For example, high-power EV charging can lead to voltage fluctuations, harmonics, power quality degradation, and distribution network stability issues [27]. Seal et al. investigated the integration of EVs as mobile energy storage units into a centralized MPC-based HEMS and analyzed the influence of uncertainties in EV arrival, departure, and trip discharge on the control performance through Monte Carlo simulations [23]. Park and Moon proposed a MADRL-based charging scheduling method for multiple EVs in a smart grid and emphasized that decentralized execution is more suitable for practical charging environments requiring fast decision-making and heterogeneous vehicle states [19]. These studies indicate that EV modeling should not simply treat EVs as conventional electrical loads, but should explicitly consider their arrival time, departure time, SoC evolution, and user requirements.

Overall, building energy management in distributed networks must simultaneously consider the economic objectives on the user side and the operational requirements on the

distribution network side. Makhadmeh et al. pointed out that the primary objectives of the smart home power scheduling problem under dynamic electricity pricing typically include reducing electricity costs, reducing the peak-to-average ratio, and improving user comfort [15]. Ali et al. summarized the architecture of HEMS in smart grids from the perspective of closed-loop control, emphasizing that HEMS involves not only the scheduling of household devices but also the coordinated operation with demand-side management, demand response, communication networks, and monitoring devices [1]. Therefore, in the low-voltage distribution network considered in this thesis, energy management must not only schedule the energy resources within individual households but also coordinate the simultaneous actions of multiple prosumers and their combined impacts on the net load.

2.2 Existing Control Methods

The objective functions, device models, user comfort requirements, prediction uncertainty, and computational complexity considered in HEMS research vary considerably [3].

Early HEMS control methods mainly use rule-based or heuristic strategies. These methods are simple to implement, only require low computational effort, and are also suitable for practical deployment. However, they lack systematic foresight regarding future electricity prices, PV generation, and load variations. Shakeri et al. proposed a HEMS architecture and control algorithm that schedules household devices according to the electricity price, battery status, and thermal appliance states. Their simulation results demonstrated a reduction in daily electricity costs, although the control logic still relied on manually designed rules [24]. Costanzo et al. formulated smart home peak-load shaving as an online scheduling problem and showed that online scheduling can reduce peak demand. However, their approach mainly focused on appliance scheduling and did not address the more complex coordination of multiple energy storage systems and multiple prosumers [7]. Wu et al. proposed a hierarchical online EMS for a residential microgrid with hybrid hydrogen–electricity storage, where a fuzzy logic controller was employed to perform energy management at different time scales. Although this method avoids dependence on prediction models, the authors also pointed out that the lack of foresight may prevent stable electricity purchasing during low-price periods [29]. Rahim et al. compared heuristic algorithms such as GA, BPSO, and ACO for home energy management controllers, demonstrating that heuristic methods can reduce electricity costs and the peak-to-average ratio, although their performance strongly depends on the algorithm design and problem formulation [21]. Sisodiya et al. applied PSO to demand response scheduling for household devices including EVs and UPS systems, demonstrating the

application of metaheuristic methods in residential energy storage and load management [26].

Different from rule-based methods, optimization-based methods can provide a more explicit representation of device constraints and economic objectives. Zhang et al. proposed a demand response and HEM system that combines machine learning, optimization, and data structure design. Household loads are classified into three categories, namely fixed, regulatable, and deferrable loads. A learning-based energy consumption model is established for HVAC systems, and the demand response strategy is subsequently generated through optimization [31]. Jin et al. proposed the Foresee HEMS based on a multi-objective MPC framework, simultaneously considering energy cost, thermal comfort, user convenience, and carbon emissions. Machine learning algorithms are employed to learn appliance models and usage patterns, thereby providing demand response services while satisfying user requirements [11]. Chen et al. proposed MPC-based appliance scheduling, where thermal and non-thermal appliances are optimized over a finite prediction horizon under time-varying retail electricity pricing, and new prediction information is continuously incorporated through a rolling optimization framework [5]. Beaudin et al. proposed a moving window algorithm that periodically updates forecasts to improve the robustness of residential energy management against forecast errors and external disturbances [4]. Zheng et al. investigated load shifting and peak reduction using building energy storage from the perspective of demand response in the residential sector, demonstrating that energy storage systems can provide demand response capability without changing the actual electricity consumption behavior [33].

The advantages of MPC are that it can explicitly consider SoC dynamics, power limits, comfort constraints, and forecast information during each rolling optimization horizon. So it is suitable for the scheduling of BESS, EVs, and PV systems. However, MPC and mixed-integer optimization also face problems including high computational complexity, strong model dependence, and reliance on optimization solvers. Mueller et al. compared rule-based control, MPC, imitation learning, and reinforcement learning for controlling multiple household BESS in a German low-voltage benchmark distribution network. This showed that model-based methods generally achieve the best objective values and the fewest constraint violations, although at the cost of significantly higher computational time. In contrast, model-free methods require less computational effort but may result in higher objective values and more distribution network constraint violations [18]. Therefore, MPC is adopted as the baseline method in this thesis.

Distributed energy system management consists of a large number of small-scale distributed energy resources. Their distributed nature, intermittent behavior, and bidirectional power flow make system operation increasingly difficult to manage using tradi-

tional centralized methods. Consequently, besides the methods mentioned above, artificial intelligence and machine learning have been also widely applied to forecasting, optimization, real-time control, intelligent demand response, and decentralized coordination to improve the adaptability, scalability, and robustness of distributed energy systems [22]. Liu et al. applied DQN and DDQN to HEMS appliance scheduling and compared their performance with PSO, demonstrating that model-free DRL can reduce electricity costs through action–reward interactions in dynamic environments, although its performance is strongly influenced by reward design and generalization capability [14]. Mocanu et al. were among the first to apply DRL to online optimization for building energy management. Using the Pecan Street dataset containing PV systems, EVs, and building appliances, they evaluated Deep Q-learning and deep policy gradient methods, showing that deep policy gradient is more suitable for online energy resource scheduling [17]. Amer et al. systematically discussed the state–action pairs, reward functions, and hyperparameters of DRL for demand response in HEMS, pointing out that the performance of DRL is highly dependent on the configuration, while reward formulation and environmental variations significantly affect the convergence speed, robustness, and scheduling performance [2]. Shuvo and Yilmaz proposed a HERS framework that incorporates resident feedback and activity information into deep reinforcement learning, making HEMS more user-centric. However, they also noted that their model does not include PV systems, energy storage, microgrids, or a multi-agent framework, leaving room for further extension [25]. Furthermore, Du et al. proposed an LLM-augmented reinforcement learning framework, in which an LLM is used during offline iterations to translate natural language user preferences into reward and state modifications, while continuously refining the policy through result analysis and policy warm-start to reduce the mismatch between the learned policy and user preferences [9].

Moreover, in multi-user and distributed scenarios, MADRL is more suitable than single-agent DRL for modeling the coordinated decision-making of multiple households or multiple EVs. Xu et al. proposed a multi-agent reinforcement learning-based data-driven HEM method, formulating hour-ahead scheduling as a finite Markov decision process. Electricity prices and PV generation are predicted using ELM, followed by multi-agent Q-learning for scheduling household appliances and EV charging [30]. Lee et al. adopted a centralized training and decentralized execution multi-agent deep actor–critic framework, allowing each household agent to rely only on local observations during execution, thereby improving both scalability and privacy preservation [12]. Wang et al. proposed a scalable MADRL framework based on multi-stage PPO and imitation learning for residential hybrid energy systems, employing centralized training and decentralized execution to improve the scalability and robustness of multi-agent systems [28]. The EV charging scheduling study by Park and Moon also adopted the CTDE framework, where

a decentralized charging policy is learned for each EV in a smart grid charging station [19]. These studies demonstrate that MADRL is capable of handling the coupling among multiple agents and the challenges associated with local observations, which is also the reason why this thesis adopts a multi-agent actor-critic framework.

At the same time, safety constraints and physical feasibility are essential considerations for the practical deployment of energy management systems. Li et al. reviewed the applications of DRL in smart grid operations and pointed out that DRL is capable of handling uncertainty, the curse of dimensionality, and the absence of accurate system models. However, challenges remain in reward design, sample efficiency, and simulation-to-reality transfer [13]. Zhang et al. proposed an interior-point policy optimization-based MADRL framework that formulates secure HEM as a constrained Markov decision process and integrates non-stationary Transformer forecasting to improve cost efficiency, comfort, and safety guarantees under uncertain environments [32]. Quakernack et al. applied DDQN-based multi-agent control to low-voltage distribution network stability by jointly controlling EV charging and battery storage system charging/discharging to mitigate voltage drops, transformer overload, and reverse power flow [20]. Based on these, this thesis will also consider the safety constraints of the distribution network.

2.3 Research Gaps

In summary, the existing literature has investigated HEMS, residential microgrids, BESS scheduling, EV charging, MPC, machine learning, DRL, and MADRL from multiple perspectives. However, several research gaps still remain.

Many HEMS studies primarily focus on the internal optimization of a single household or a single building, with objectives such as reducing electricity costs, improving user comfort, or increasing PV self-consumption [34, 3, 11], but they pay limited attention to the network coupling effects arising when multiple prosumers are connected to the same low-voltage distribution network. In this thesis, the battery charging/discharging, PV curtailment, and EV charging decisions of multiple prosumers jointly modify the feeder net load and subsequently affect the bus voltage, line loading, and transformer loading through power flow calculations. Therefore, the conclusions obtained from single-household HEMS cannot be directly extended to multi-prosumer distribution network energy management.

Traditional optimization and MPC methods can explicitly represent device constraints and future forecast information, but their performance strongly depends on accurate system models, forecast quality, and optimization efficiency. MPC has demonstrated

strong performance in residential appliance scheduling, battery peak shaving, and EV-as-mobile-storage problems [5, 8, 23]. The comparative study by Mueller et al. also showed that model-based methods generally achieve better objective values and fewer constraint violations [18]. However, when multiple prosumers, BESSs, PV systems, EVs, dynamic electricity prices, and distribution network constraints are simultaneously considered, both the optimization problem size and computational complexity increase significantly. In particular, MISOCP and mixed-integer formulations become more difficult to deploy in real time due to binary variables and nonlinear power flow approximations.

Furthermore, although DRL and MADRL are suitable for handling uncertainty and online decision-making, training stability, reward design, and constraint satisfaction remain key challenges. Existing DRL-based HEMS studies have shown that DQN, DDQN, deep policy gradient, and multi-agent Q-learning can reduce residential energy costs or improve demand response performance [14, 17, 30]. However, Amer et al. pointed out that the performance of DRL is highly dependent on the reward function and hyperparameters [2]. Li et al. also emphasized that reward design, exploration, and safety remain important limitations in DRL applications for smart grids [13]. Therefore, in this thesis, different reward designs and constraint handling methods are investigated to analyze their effects on economic performance, EV feasibility, and distribution network safety.

Although existing multi-agent methods are used to support decentralized execution for multiple users or multiple EVs, studies that simultaneously consider residential BESS, PV utilization, EV departure SoC, forecasted electricity price, load and PV generation, together with distribution network operational indicators, remain limited. Xu et al. and Lee et al. demonstrated the potential of MARL for HEM and demand-side scheduling [30, 12]. Park and Moon showed that the CTDE framework is well suited for multi-EV charging scheduling [19], while Wang et al. demonstrated that MADRL can improve the scalability and training efficiency of residential hybrid energy systems [28]. However, these studies mainly focus on charging stations, building thermal comfort, or hybrid energy systems, so research on low-voltage distribution networks with multiple prosumers, as considered in this thesis, remains limited.

In addition, distribution network safety constraints are often simplified in the HEMS and DRL literature. EVs, battery storage systems, and PV systems affect voltage drop, reverse power flow, and transformer loading, while reinforcement learning has been applied to distribution network stability control [20]. Zhang et al. also demonstrated in their secure HEM framework that an unrestricted action space may violate physical constraints, thereby motivating the development of safe MADRL methods [32]. However, the objective of this thesis is not to propose an entirely new safety projection theory.

Instead, this work compares different control methods for multi-prosumer EV–BESS–PV scheduling and analyzes the effects of soft constraints, hard projection, and emergency EV constraint handling on economic performance, EV feasibility, and distribution network safety.

Based on the above research gaps, the research objective of this thesis can be summarized as follows. A multi-prosumer energy management environment is established for a low-voltage distribution network, incorporating household loads, PV generation, stationary BESS, residential EV charging, electricity prices, and power-flow feedback. Rule-based control, Local MPC, ADMM-MPC, and MADRL controllers are compared. Particular emphasis is placed on the soft-constraint and hard-projection approaches for handling the EV departure SoC requirement, together with their impacts on total cost, battery arbitrage, EV charging cost, departure SoC, and distribution network safety. This thesis aims to bridge the gap between single-household HEMS, optimization-based control methods, and MADRL approaches, thereby providing more comprehensive experimental evidence for the coordinated control of EVs and energy storage systems in multi-prosumer scenarios.

3 System Modelling and Problem Formulation

In this chapter, the multi-prosumer building energy management problem investigated in this thesis is defined, and the physical models and mathematical formulations involved in the environment are presented. The system under study consists of three prosumers connected to a low-voltage distribution network. Each prosumer is equipped with local electrical loads, PV generation, a stationary battery energy storage system, and residential EV charging demand. At each discrete time step, the control system determines the battery charging/discharging power, PV utilization, and EV charging power to minimize the total electricity purchasing cost while satisfying both the physical constraints of different devices and the safety constraints of the distribution network.

This chapter focuses only on the system modeling and problem formulation, including the system constraints and the optimization objective function. The details of the control methodology, such as how MADRL actions are generated, how the EV control strategies are implemented at the control layer, and how the grid safety projection modifies the control actions, are discussed in Chapter 4.

3.1 Overall System

The system investigated in this thesis addresses the energy management problem of a low-voltage distribution network integrated with distributed energy resources. The system consists of a low-voltage distribution network, 3 prosumer agents, household loads, PV generation, a stationary battery energy storage system, EV charging, electricity price signals, and a power flow calculation module. Each prosumer acts not only as an electricity consumer but also as an electricity producer and the owner of dispatchable energy resources. The primary objective of the system is to coordinate the resources of the three prosumers to minimize the overall operating cost while satisfying both the device operational constraints and the distribution network safety constraints.

The distribution network topology is adopted from SimBench, with the fixed network

configuration:

$$1\text{-LV-rural1--0-sw.}$$

The system consists of $N = 3$ prosumers connected to the distribution network buses

$$\mathcal{B} = \{12, 4, 2\}.$$

Let $i \in \mathcal{I} = \{1, 2, 3\}$ denote the prosumer index, and $t \in \mathcal{T} = \{0, \dots, T-1\}$ represent the discrete time step within an episode. The simulation uses a 15-minute resolution, so a day is represented as:

$$\Delta t = 0.25 \text{ h}, \quad T = 96.$$

One episode corresponds to the operation of one whole day. During training, multiple consecutive days can be concatenated to form a longer training window.

The local power profile of each prosumer i consists of external inputs and three kinds of controllable or semi-controllable energy resources. The external inputs include the aggregated electrical demand $L_{i,t}$, the available PV generation $G_{i,t}$, and the electricity price p_t . The aggregated demand may contain several exogenous load components, whose general formulation is introduced in Section 3.2; the component selection and numerical settings used in the experiments are reported in Chapter 5. The controllable resources include the charging and discharging power of the stationary battery $P_{i,t}^b$, the utilized PV generation $G_{i,t}^{use}$, and the EV charging power $P_{i,t}^{ev}$. Accordingly, the electricity-related variables of each prosumer can be expressed as

$$\{L_{i,t}, G_{i,t}, P_{i,t}^b, P_{i,t}^{ev}\},$$

where $L_{i,t}$ denotes the total local electrical load, $G_{i,t}$ denotes the available PV generation, $P_{i,t}^b$ denotes the stationary battery power, and $P_{i,t}^{ev}$ denotes the EV charging power. In this thesis, $P_{i,t}^b > 0$ indicates battery charging, whereas $P_{i,t}^b < 0$ indicates battery discharging.

$$-P_i^{b,\max} \leq P_{i,t}^b \leq P_i^{b,\max}.$$

Since only EV charging is considered in this thesis, while vehicle-to-grid is not allowed, the EV charging power satisfies

$$P_{i,t}^{ev} \geq 0.$$

At each time step, the power of all devices within a prosumer is aggregated to determine the net load injected into the distribution network:

$$P_{i,t}^{net} = L_{i,t} - G_{i,t}^{use} + P_{i,t}^b + P_{i,t}^{ev}.$$

When $P_{i,t}^{net} > 0$, the prosumer imports electricity from the distribution grid; when $P_{i,t}^{net} < 0$, the prosumer exports power back to the grid. The net loads of the three prosumers are injected into the low-voltage distribution network model, where power flow analysis is performed using pandapower. The power flow calculation returns the bus voltages, line loading ratios, and transformer loading ratios, which are subsequently used to determine whether the control actions satisfy the distribution network safety constraints.

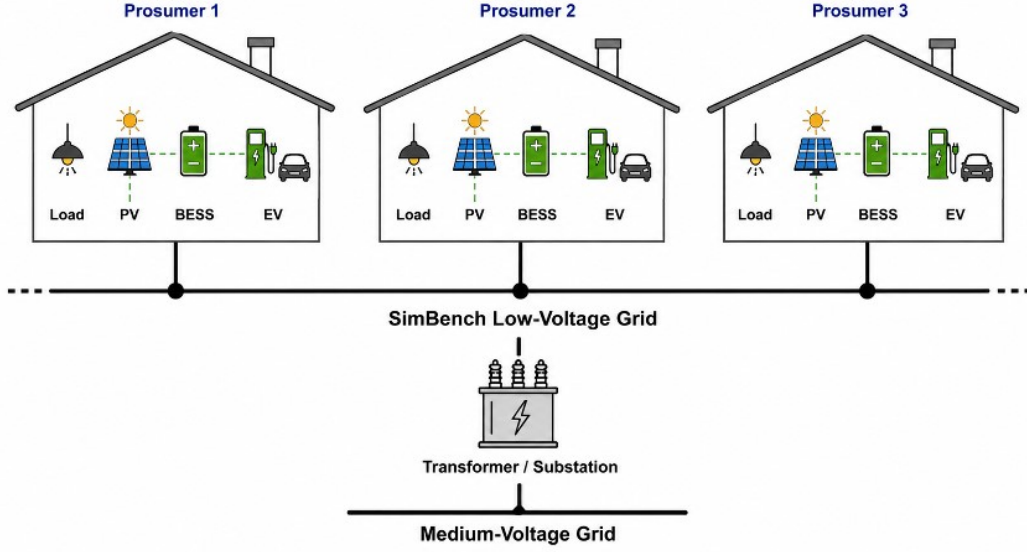


Figure 3.1: Overall architecture of the system investigated in this thesis.

Figure 3.1 illustrates the overall architecture of the system. The household load, PV generation, and electricity price are treated as time-series inputs from the dataset. The stationary battery and EV charging serve as the dispatchable resources on the prosumer side. The grid power flow calculation module maps the net load of each prosumer to the corresponding network operating states, including bus voltages and equipment loading conditions. Accordingly, the problem addressed in this thesis can be formulated as follows: at each discrete time step, the control system determines $P_{i,t}^b$, $G_{i,t}^{use}$, and $P_{i,t}^{ev}$ for each prosumer based on the load demand, PV generation, electricity price, and the state of charge of the energy storage devices, with the objective of minimizing the total cost while ensuring the safe operation of the stationary battery, the EV, and the distribution network.

3.2 Load of the Prosumers

The exogenous electrical demand of each prosumer can consist of multiple load components. Let $\mathcal{C}^L$ denote the set of load components, and let $L_{c,i,t}^{raw}$ represent the unscaled demand of component $c \in \mathcal{C}^L$ for prosumer i at time step t . An agent-specific scaling factor s_i^L maps the original profile to the demand level considered in the simulation. The aggregated prosumer load is therefore

$$L_{i,t} = s_i^L \sum_{c \in \mathcal{C}^L} L_{c,i,t}^{raw}.$$

This formulation allows different electrical demand components and prosumer-specific demand levels to be represented without changing the subsequent battery, EV, PV, or grid equations. The experimental choices for $\mathcal{C}^L$, the selected prosumer profiles, and s_i^L are specified in Chapter 5. The resulting $L_{i,t}$ represents only exogenous demand; PV generation, stationary-battery power, and EV charging are incorporated separately in the net-load equation.

3.3 PV Modelling

The available PV power is obtained from the data file

`datasets/prosumer/pv_reference.csv`.

In the current configuration, the reference column `ref_south` is used. The same reference PV profile is assigned to all three prosumers and scaled by an agent-specific scaling factor s_i^G ,

$$s_1^G = s_2^G = s_3^G = 1.$$

Accordingly, the available PV generation is given by

$$G_{i,t} = s_i^G G_t^{ref}.$$

The controller does not directly modify the available PV generation $G_{i,t}$; instead, it determines the utilized PV power $G_{i,t}^{use}$. Let $u_{i,t}^{pv} \in [0, 1]$ denote the PV utilization factor. Then,

$$\begin{aligned} G_{i,t}^{use} &= u_{i,t}^{pv} G_{i,t}, \\ G_{i,t}^{curt} &= G_{i,t} - G_{i,t}^{use}. \end{aligned}$$

$G_{i,t}^{curt}$ denotes the curtailed PV power. In the current model, no explicit penalty is assigned to PV curtailment. Therefore, PV curtailment primarily affects the system performance indirectly by changing the net load and the operating state of the distribution network.

3.4 Stationary Battery Energy Storage Model

Each prosumer is equipped with a stationary battery. Let C_i^b denote the battery capacity, and let $SOC_{i,t}^b$ denote the battery state of charge (SoC) at the beginning of time step t . The default parameters are

$$C_i^b = 100 \text{ kWh}, \quad SOC_{\min}^b = 0.05, \quad SOC_{\max}^b = 0.95.$$

The battery charging and discharging efficiency is

$$\eta_b = 0.95.$$

Maximum charging and discharging powers are determined by the capacity and maximum charging rate:

$$P_i^{b,\max} = r_{\max}^b C_i^b,$$

where the default $r_{\max}^b = 0.5$, thus a 100 kWh battery corresponds to

$$P_i^{b,\max} = 50 \text{ kW}.$$

The battery power constraints are

$$-P_i^{b,\max} \leq P_{i,t}^b \leq P_i^{b,\max}.$$

The battery SoC dynamics are determined by the charging and discharging directions as mentioned before. The SoC is updated as

$$SOC_{i,t+1}^b = \begin{cases} SOC_{i,t}^b + \frac{P_{i,t}^b \Delta t \eta_b}{C_i^b}, & P_{i,t}^b \geq 0, \\ SOC_{i,t}^b + \frac{P_{i,t}^b \Delta t}{\eta_b C_i^b}, & P_{i,t}^b < 0. \end{cases}$$

subject to the boundary constraint

$$SOC_{\min}^b \leq SOC_{i,t+1}^b \leq SOC_{\max}^b.$$

This formulation shows the charging and discharging efficiency losses. During charging, only a fraction η_b of the input electrical energy is stored in the battery. During discharging, a larger amount of internal battery energy must be consumed to deliver the same amount of electrical energy to the external system.

3.5 EV Modelling

Each prosumer is equipped with one residential EV. In the distribution network model, the EV is represented as a controllable charging load characterized by a connection time window, a charging power limit, a battery capacity, and a required departure SoC. In this thesis, vehicle-to-grid operation is not considered. Therefore, the EV can only draw electrical energy while connected to the residential charger. Let C_i^{ev} denote the EV battery capacity, and let $SOC_{i,t}^{ev}$ denote SoC of the EV at the beginning of time step t . The default parameters are

$$C_i^{ev} = 60 \text{ kWh}, \quad SOC_{\min}^{ev} = 0.10, \quad SOC_{\max}^{ev} = 0.95.$$

The maximum power of the residential charger is

$$P_i^{ev,\max} = 11 \text{ kW},$$

and the EV charging efficiency is

$$\eta_{ev} = 0.95.$$

Since the simulation time step is $\Delta t = 0.25 \text{ h}$, the maximum amount of energy that can be charged during a single 15-minute time step is

$$E_{i,\Delta t}^{ev,\max} = P_i^{ev,\max} \Delta t \eta_{ev}.$$

3.5.1 EV Connection Window

EV's connection status is determined by the intraday time steps. The current assumption is that the vehicle arrives home at 18:00 and connects to the residential charger, and departs at 07:00 the next day. Since there are 96 fifteen-minute time steps in a day, the arrival and departure times are

$$t^{arr} = 72, \quad t^{dep} = 28.$$

Let $t_d = t \bmod T$ denote the intraday time step. Since $t^{arr} > t^{dep}$, the connection window spans midnight, and thus a complete parking window consists of the time steps $72, \dots, 95$ from the evening of the current day and the time steps $0, \dots, 27$ from the early morning of the next day:

$$\mathcal{T}^{ev} = \{t : t_d \geq t^{arr}\} \cup \{t : t_d < t^{dep}\}.$$

The connecting status of the EV is represented by the variable A_t^{ev} :

$$A_t^{ev} = \begin{cases} 1, & t_d \geq t^{arr} \text{ or } t_d < t^{dep}, \\ 0, & \text{otherwise.} \end{cases}$$

When $A_t^{ev} = 1$, EV is at home and can be charged; when $A_t^{ev} = 0$, the vehicle is away, and the charging power of EV must be zero.

3.5.2 EV arrival SoC and Charging Power Constraints

Upon arrival at home, the EV SoC is initialized to the arrival SoC:

$$SOC_{i,t}^{ev} = SOC_{arr}^{ev} = 0.30, \quad t_d = t^{arr}.$$

The EV charging power must satisfy both the connection status and the charger power limit:

$$0 \leq P_{i,t}^{ev} \leq A_t^{ev} P_i^{ev,max}.$$

Furthermore, the EV SoC must not exceed its upper limit. Expressing the remaining SoC headroom as an equivalent charging power yields an additional upper bound determined by the remaining battery capacity:

$$P_{i,t}^{ev} \leq \frac{(SOC_{max}^{ev} - SOC_{i,t}^{ev})C_i^{ev}}{\Delta t \eta_{ev}}.$$

Therefore, the physically feasible upper bound on the EV charging power can be written as

$$0 \leq P_{i,t}^{ev} \leq \min \left(A_t^{ev} P_i^{ev,max}, \frac{(SOC_{max}^{ev} - SOC_{i,t}^{ev})C_i^{ev}}{\Delta t \eta_{ev}} \right).$$

When the EV is not connected, i.e., $A_t^{ev} = 0$, the above formulation automatically yields $P_{i,t}^{ev} = 0$. As the EV approaches SOC_{max}^{ev} , the SoC headroom constraint progressively limits the charging power, thereby preventing the battery from being charged beyond its physical capacity.

3.5.3 EV SoC Status

The change of the EV SoC is determined by the actual charging power:

$$SOC_{i,t+1}^{ev} = SOC_{i,t}^{ev} + \frac{P_{i,t}^{ev} \Delta t \eta_{ev}}{C_i^{ev}}.$$

Since V2G is not considered in this thesis, $P_{i,t}^{ev} \geq 0$. Consequently, the EV SoC is monotonically non-decreasing throughout the connection window. During time steps when the EV is not connected and no arrival SoC reset occurs, the EV does not exchange energy with the residential charging system. At all time steps, the EV SoC must satisfy the following physical constraints:

$$SOC_{min}^{ev} \leq SOC_{i,t}^{ev} \leq SOC_{max}^{ev}.$$

3.5.4 EV Departure SoC Requirement

Another physical requirement for the EV is to have sufficient charge when departing from home. The default departure target SoC is set to

$$SOC_{req}^{ev} = 0.90.$$

Therefore, for each prosumer i , the SoC requirement at departure time is

$$SOC_{i,dep}^{ev} \geq SOC_{req}^{ev}.$$

Since the connection window spans midnight, this constraint links the charging decisions made after the EV arrives home in the evening to its availability for use on the following morning. Equivalently, the minimum increase in battery energy required to charge the EV from the arrival SoC to the required departure SoC is

$$\Delta E_i^{ev,req} = \max(0, SOC_{req}^{ev} - SOC_{arr}^{ev}) C_i^{ev}.$$

Considering the charging efficiency, the minimum amount of energy that must be supplied from the grid is

$$E_i^{ev,grid,req} = \frac{\Delta E_i^{ev,req}}{\eta_{ev}}.$$

Under the default parameter settings, charging the EV from 0.30 to 0.90 requires an increase of $0.60 \times 60 = 36$ kWh in the battery energy. Accounting for the charging efficiency $\eta_{ev} = 0.95$, the corresponding energy that must be drawn from the grid is approximately

$$\frac{36}{0.95} \approx 37.9 \text{ kWh}.$$

3.6 Electricity Price Model

The electricity price data are obtained from

$$\text{datasets/prosumer/price.csv}.$$

Let p_t denote the wholesale electricity price, expressed in EUR/kWh. The electricity import price used in both the environment and the optimizer is defined as

$$\tilde{p}_t = p_t + \mu^{imp},$$

where μ^{imp} denotes the electricity import price markup. In the current default configuration,

$$\mu^{imp} = 0.$$

Therefore, under the default setting $\tilde{p}_t = p_t$

3.7 Net Load, Power Flow, and Grid Safety

After integrating the load, PV, battery, and EV, the net load injected by prosumer i into the distribution grid at time step t is given by

$$P_{i,t}^{net} = L_{i,t} - G_{i,t}^{use} + P_{i,t}^b + P_{i,t}^{ev}.$$

As mentioned before, when $P_{i,t}^{net} \geq 0$, the prosumer draws power from the grid; when $P_{i,t}^{net} < 0$, it injects power into the grid.

In the pandapower power flow model, positive net load is written to the load table, and negative net load is written to the sgen table:

$$P_{i,t}^{load} = \frac{\max(P_{i,t}^{net}, 0)}{1000},$$

$$P_{i,t}^{sgen} = \frac{\max(-P_{i,t}^{net}, 0)}{1000}.$$

The factor 1000 is used to convert kW to MW. Subsequently, the environment performs AC power flow calculations to obtain the voltages at each bus, line loading rates, and transformer loading rates. In other words, grid safety is not determined by simple algebraic power balance but is jointly decided by the distribution grid topology, line impedances, transformer parameters, and the net injections from each prosumer.

3.7.1 Voltage Constraints

Let $V_{b,t}$ denotes the voltage magnitude at bus b at time step t , expressed in per unit. The voltage safety range is given by

$$V_{\min} = 0.95, \quad V_{\max} = 1.05.$$

Therefore, all buses should satisfy

$$V_{\min} \leq V_{b,t} \leq V_{\max}, \quad b \in \mathcal{B}^{grid}.$$

For any bus b , the voltage violation can be written as

$$v_{b,t} = \max(0, V_{\min} - V_{b,t}) + \max(0, V_{b,t} - V_{\max}).$$

The total voltage violation across the network is measured as

$$\psi_t^V = \sum_{b \in \mathcal{B}^{grid}} v_{b,t}^2.$$

When all bus voltages are within the range $[0.95, 1.05]$, $v_{b,t} = 0$ and $\psi_t^V = 0$. If certain buses fall below or exceed the safety limits due to high load, centralized EV charging, or high PV generation, then ψ_t^V will become positive.

3.7.2 Line Loading Constraints

The safety of lines is measured by their loading percentage. Let $\lambda_{\ell,t}$ denote the loading percentage of line ℓ at time step t . Physically, $\lambda_{\ell,t}$ represents the ratio of the line's apparent power or current to its rated capacity. The current safety limit is

$$\lambda_{\max}^{line} = 100.$$

Therefore, all lines should satisfy

$$\lambda_{\ell,t} \leq \lambda_{\max}^{line}, \quad \ell \in \mathcal{L}.$$

The line violation measure is expressed as

$$\psi_t^{line} = \sum_{\ell} \left(\frac{\max(0, \lambda_{\ell,t} - \lambda_{\max}^{line})}{100} \right)^2,$$

where dividing by 100 is to convert the percentage violation into a relative measure. When all line loadings do not exceed 100%, $\psi_t^{line} = 0$.

3.7.3 Transformer Loading Constraints

Transformer safety is similarly measured by the loading percentage. Let $\lambda_{m,t}^{tr}$ denote the loading percentage of transformer m at time step t , and let $\lambda_{\max}^{tr}$ denote the transformer loading limit. Therefore, all transformers should satisfy

$$\lambda_{m,t}^{tr} \leq \lambda_{\max}^{tr}, \quad m \in \mathcal{M}^{tr}.$$

The corresponding transformer violation measure is

$$\psi_t^{tr} = \sum_m \left(\frac{\max(0, \lambda_{m,t}^{tr} - \lambda_{\max}^{tr})}{100} \right)^2.$$

Transformer constraints are particularly sensitive to the system's total net load. When multiple prosumers charge their EVs or batteries simultaneously, the power demand at the feeder head may increase, thereby raising transformer loading; when PV generation is high and local load is low, reverse power flow may also affect transformer and line states.

In summary, grid safety constraints can be summarized as

$$\begin{aligned} V_{\min} &\leq V_{b,t} \leq V_{\max}, & b &\in \mathcal{B}^{grid}, \\ \lambda_{\ell,t} &\leq \lambda_{\max}^{line}, & \ell &\in \mathcal{L}, \\ \lambda_{m,t}^{tr} &\leq \lambda_{\max}^{tr}, & m &\in \mathcal{M}^{tr}. \end{aligned}$$

These safety metrics do not alter the physical power flow results themselves, but are used to evaluate whether given control actions lead to voltage violations, line overloads, or transformer overloads.

3.8 Optimization Objective

The objective of this study is to minimize the total electricity cost of the system while satisfying both the device constraints and the distribution network constraints. For prosumer i , the economic cost associated with the stationary battery at time step t is given by

$$C_{i,t}^b = \tilde{p}_t \Delta t P_{i,t}^b.$$

Since $P_{i,t}^b > 0$ denotes battery charging, $C_{i,t}^b$ represents a positive charging cost. Conversely, when $P_{i,t}^b < 0$, the battery is discharging, resulting in a negative value of $C_{i,t}^b$, which corresponds to a reduction in the electricity purchasing cost or, equivalently, revenue from battery discharge.

The EV charging cost is given by

$$C_{i,t}^{ev} = \tilde{p}_t \Delta t P_{i,t}^{ev}.$$

In the current model, PV curtailment does not have an explicit economic price, thus

$$C_{i,t}^{pv} = 0.$$

Therefore, the total economic cost of the single-step system is

$$C_t = \sum_{i=1}^N \left(C_{i,t}^b + C_{i,t}^{ev} + C_{i,t}^{pv} \right).$$

In an episode, the objective function can be formulated as

$$\begin{aligned}
 & \min_{\{P^b, G^{use}, P^{ev}\}} \sum_{t=0}^{T-1} \sum_{i=1}^N \tilde{p}_t \Delta t \left(P_{i,t}^b + P_{i,t}^{ev} \right) \\
 & \text{s.t.} \quad L_{i,t} = s_i^L L_{i,t}^{hh,raw}, \\
 & \quad 0 \leq G_{i,t}^{use} \leq G_{i,t}, \\
 & \quad -P_i^{b,max} \leq P_{i,t}^b \leq P_i^{b,max}, \\
 & \quad SOC_{min}^b \leq SOC_{i,t}^b \leq SOC_{max}^b, \\
 & \quad 0 \leq P_{i,t}^{ev} \leq A_t^{ev} P_i^{ev,max}, \\
 & \quad SOC_{min}^{ev} \leq SOC_{i,t}^{ev} \leq SOC_{max}^{ev}, \\
 & \quad SOC_{i,t}^{ev,dep} \geq SOC_{req}^{ev}, \\
 & \quad P_{i,t}^{net} = L_{i,t} - G_{i,t}^{use} + P_{i,t}^b + P_{i,t}^{ev}, \\
 & \quad V_{min} \leq V_{b,t} \leq V_{max}, \\
 & \quad \lambda_{\ell,t} \leq \lambda_{max}^{line}, \\
 & \quad \lambda_{m,t}^{tr} \leq \lambda_{max}^{tr}.
 \end{aligned}$$

The problem has a multi-time-step coupling characteristic: the battery SoC and EV SoC make the current actions affect the future feasible region; PV, load, and price exhibit obvious time variations; and the distribution network flow causes different prosumer actions to be coupled through voltage and line loading.

4 Methodology

This chapter presents the prediction, control, safety correction, and training methods used in this thesis. Based on the system model introduced in Chapter 3, this chapter further describes how these models are implemented in the code. The overall framework consists of an LSTM-based forecasting module, a pandapower-based simulation environment, a MADRL controller, a safety action projection mechanism, and EV constraint handling mechanism.

4.1 Overall Framework

The overall workflow can be summarized as follows. First, the historical data are processed into time-series sequences of load demand, PV generation, and electricity prices. The LSTM predictor then generates forecasts over a future prediction horizon. The environment constructs the observation by combining the current system state with the predicted sequences. Based on its own observation, each MADRL actor outputs a control action. The generated actions subsequently undergo local feasibility checks and, if enabled, a safety action projection procedure. The environment then updates the battery SoC, EV SoC, and the net load of each prosumer according to the final control actions. Next, pandapower performs the power flow calculation using the updated net loads. Finally, the environment computes the reward based on the electricity cost, EV charging requirements, battery behavior, and the distribution network operating condition, and returns the reward to the agents for training.

Let x_t denote the external input data at the current time step, including load $L_{i,t}$, PV generation $G_{i,t}$, and electricity price p_t . Let $\hat{x}_{t:t+H-1}$ denote the prediction window of length H . The process within a time step is given by

$$x_t, \hat{x}_{t:t+H-1}, SOC_t^b, SOC_t^{ev} \rightarrow o_t \rightarrow a_t \rightarrow \bar{a}_t \rightarrow s_{t+1}, r_t.$$

where o_t is the observation, a_t is the actor's raw action, $\bar{a}_t$ is the executed action after feasibility checks or projection, s_{t+1} is the environment's next state, and r_t is the immediate reward for each agent.

The overall framework of this work includes the following steps:

1. Read historical household load, PV generation, and electricity price.
2. Use LSTM model to generate future prediction sequences for electricity price, load, and PV generation.
3. The environment constructs the observation based on the current SoC, EV connection status, current exogenous variables, and the prediction window.
4. Each agent's actor outputs three action components: battery action, PV action, and EV action.
5. The battery action is first mapped to the feasible power range of SoC; in projection-based variants, battery power and PV curtailment are further subjected to grid safety projection.
6. The environment updates the battery SoC, EV SoC, PV utilization, and net load based on the final actions.
7. pandapower writes the prosumer net load to the low-voltage distribution network and performs power flow calculation.
8. The environment calculates the reward based on economic terms, EV-related terms, battery-related terms, and grid safety terms.
9. The transition is written to the replay buffer and is used to update the actor and critic networks.

Figure 4.1 illustrates the main workflow of this work.

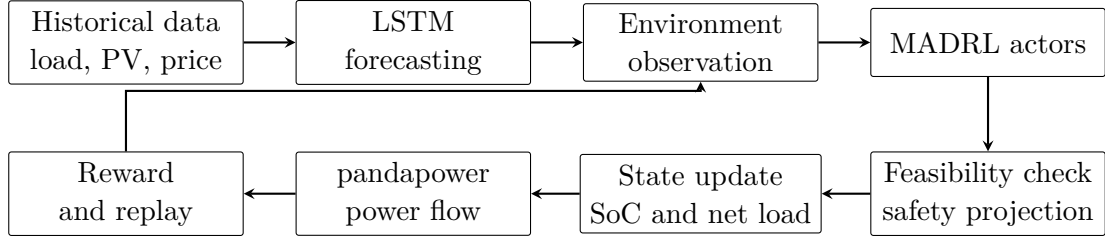


Figure 4.1: The work flow of the MADRL energy management framework.

4.2 LSTM Forecasting

This thesis employs an LSTM forecasting module to provide future-horizon information to the controller. The forecasting targets include the electricity price, prosumer load, and PV generation. The prediction results determine the amount of future information available to the controller and can influence the scheduling performance of the stationary battery, PV curtailment, and EV charging.

For any scalar sequence y_t , LSTM uses a historical window of length h as input and outputs a predictive window of length H . The current project configuration is

$$h = 192, \quad H = 49.$$

Since the simulation time step is 0.25 h, a historical window of 192 time steps corresponds to the previous 48 hours, while a prediction window of 49 time steps includes the current time step and approximately the following 12 hours. The training sample can be formulated as

$$\begin{aligned} X_k &= (y_{k-h}, y_{k-h+1}, \dots, y_{k-1}), \\ Y_k &= (y_k, y_{k+1}, \dots, y_{k+H-1}). \end{aligned}$$

The LSTM learns the mapping

$$\hat{Y}_k = f_\theta(X_k, \phi_k),$$

where ϕ_k denotes the optional temporal features, such as the hour of the day, the day of the week, or annual cyclic features.

In this work, the electricity price is predicted using a separate LSTM model; the PV generation is predicted using a shared model trained on reference PV sequences; and the load prediction is organized by prosumer and load components.

The predicted sequences used in the environment are

$$\hat{p}_{t:t+H-1}, \quad \hat{L}_{i,t:t+H-1}, \quad \hat{G}_{i,t:t+H-1}.$$

In the perfect forecast mode, these sequences are directly obtained from the ground-truth future values. In the LSTM forecast mode, they are generated by the trained LSTM predictor. All the MADRL controller is trained using the LSTM forecast mode.

4.3 MADRL Observation

Observation of MADRL is divided into three components: the local observation, the forecast sequence observation, and the safety observation. This design follows the centralized training and decentralized execution (CTDE) paradigm. Specifically, each actor

makes decentralized decisions based on its own local state and the forecast horizon, while the critic has access to the joint information of all agents during the training process.

4.3.1 Local Observation

The local observation includes the physical states of each agent at the current time step:

$$o_{i,t}^{local} = [SOC_{i,t}^b, SOC_{i,t}^{ev}, A_t^{ev}, L_{i,t}, G_{i,t}, p_t].$$

In the MADRL actor, the local information is further transformed into `madr1_local`. This feature includes intraday and annual cyclic time features, normalized battery SoC, normalized EV SoC, and EV availability indicators:

$$o_{i,t}^{madr1} = [\sin(2\pi \text{hour}_t / 24), \cos(2\pi \text{hour}_t / 24), \\ \sin(2\pi \text{doy}_t / 365.25), \cos(2\pi \text{doy}_t / 365.25), \\ \bar{SOC}_{i,t}^b, \bar{SOC}_{i,t}^{ev}, A_t^{ev}].$$

This representation enables the actor not only to perceive the current states of the devices but also to identify the current temporal position within both the daily and annual cycles.

4.3.2 Forecast Sequence Observation

The forecast sequence observation describes the future trajectories of the electricity price, load demand, and PV generation. In the environment, the sequence can be expressed as

$$o_{i,t}^{seq} = [\hat{p}_{t:t+H-1}, \hat{L}_{i,t:t+H-1}, \hat{G}_{i,t:t+H-1}].$$

In practice, the MADRL actor uses normalized or relative representations of these forecast sequences:

$$\bar{p}_{t:t+H-1}, \quad \Delta \bar{p}_{t:t+H-1}, \quad \bar{L}_{i,t:t+H-1}, \quad \bar{G}_{i,t:t+H-1}.$$

where $\bar{p}$ denotes the relative electricity price within the prediction horizon, $\Delta \bar{p}$ represents the price variation intensity over the prediction window, and $\bar{L}$ and $\bar{G}$ denote the normalized load and PV generation forecasts, respectively. Physically, these features enable the actor to schedule the stationary battery and EV charging according to anticipated periods of high or low electricity prices, while also determining whether battery capacity should be reserved or utilized based on the forecasted PV generation.

4.3.3 Safety Observation

The safety observation is used for feasibility checking and the safety projection procedure. It contains the physical parameters required to determine the device operating limits and perform the grid safety projection:

$$o_{i,t}^{safe} = [SOC_{i,t}^b, SOC_{i,t}^{ev}, A_t^{ev}, L_{i,t}, G_{i,t}, C_i^b, P_i^{b,max}, C_i^{ev}, P_i^{ev,max}].$$

Here, the battery capacity, battery maximum charging and discharging power, EV battery capacity, and EV maximum charging power directly determine the physically feasible action space. For example, the battery SoC together with $P_i^{b,max}$ determines the maximum amount of energy that can be charged into or discharged from the battery at the current time step. Similarly, the EV connection status together with $P_i^{ev,max}$ determines whether the EV is currently available for charging and, if so, the maximum allowable charging power.

4.4 MADRL Action Design

Each agent's actor outputs a three-dimensional action vector:

$$a_{i,t} = [a_{i,t}^b, a_{i,t}^{pv}, a_{i,t}^{ev}] \in [-1, 1]^3.$$

The three action components correspond to the stationary battery, PV utilization, and EV charging, respectively.

4.4.1 Battery Action Mapping

The raw battery action generated by the actor is not executed directly as the physical battery power. Instead, it is first mapped to the feasible power range determined by the current battery SoC. Given the current SoC, the battery cannot be charged beyond $SOC_{\max}^b$ or discharged below $SOC_{\min}^b$ within a single time step. Therefore, the feasible battery power limits at the current time step are given by

$$P_{i,t}^{b,low} = -\min\left(P_i^{b,max}, \frac{(SOC_{i,t}^b - SOC_{\min}^b)C_i^b\eta_b}{\Delta t}\right),$$

$$P_{i,t}^{b,up} = \min\left(P_i^{b,max}, \frac{(SOC_{\max}^b - SOC_{i,t}^b)C_i^b}{\eta_b\Delta t}\right).$$

The raw action $a_{i,t}^b \in [-1, 1]$ is linearly mapped to this interval:

$$P_{i,t}^b = P_{i,t}^{b,low} + \frac{a_{i,t}^b + 1}{2} (P_{i,t}^{b,up} - P_{i,t}^{b,low}).$$

This ensures that even if the actor outputs a boundary value, the executed action does not directly cause the battery SoC to violate its limits.

4.4.2 PV Action Mapping

The PV action represents the PV utilization factor. It is defined as

$$u_{i,t}^{pv} = \min \left(1, \max \left(0, \frac{a_{i,t}^{pv} + 1}{2} \right) \right).$$

The corresponding utilized PV generation and curtailed PV power are given by

$$G_{i,t}^{use} = u_{i,t}^{pv} G_{i,t},$$

$$G_{i,t}^{curt} = G_{i,t} - G_{i,t}^{use}.$$

When $a_{i,t}^{pv} = 1$, the agent uses all available PV; when $a_{i,t}^{pv} = -1$, the PV is completely curtailed. The physical meaning of the PV action is to reduce reverse power flow or voltage violation risks through controllable curtailment.

4.4.3 EV Action Mapping

The EV action represents the utilization factor of the residential charger. It is defined as

$$u_{i,t}^{ev} = \min \left(1, \max \left(0, \frac{a_{i,t}^{ev} + 1}{2} \right) \right).$$

Then, EV charging power requested by the actor is

$$P_{i,t}^{ev,rl} = A_t^{ev} u_{i,t}^{ev} P_i^{ev,max}.$$

If the EV is not connected, $A_t^{ev} = 0$, the EV charging power is always zero regardless of the action produced by the actor. Subsequently, the environment determines the final executed charging power $P_{i,t}^{ev}$ by considering the EV SoC headroom as well as the hard projection or emergency charging mechanism, if enabled.

4.5 Reward Function Design

The reward function combines the economic objective, EV mobility requirements, battery behavior, and grid safety. For each agent i , the total reward consists of multiple components. Positive terms represent benefits or incentives, whereas negative terms represent costs or constraint violations.

4.5.1 EV Charging Cost

EV charging draws electrical energy from either the grid or local energy resources. Its economic cost is determined by the actual charging power and the electricity price. Let

$$\tilde{p}_t = p_t + \mu^{imp}$$

denote the electricity price. The basic EV charging cost is then given by

$$\mathcal{C}_{i,t}^{ev} = w_{cost}^{ev} P_{i,t}^{ev} \tilde{p}_t \Delta t.$$

The corresponding reward term is

$$r_{i,t}^{ev,cost} = -\mathcal{C}_{i,t}^{ev}.$$

EV charging during high-price periods incurs higher costs, while charging during low-price periods is more economical.

4.5.2 EV Cost-Weight Modelling

Experimental results show that using only the EV charging cost as the reward encourages overly conservative EV charging behavior, even during periods of low electricity prices. The parameter w_{cost}^{ev} is therefore introduced to adjust the relative importance of the EV charging cost in the MADRL reward function. In the current implementation, the soft and hard modes use the configuration parameter `ev_charging_cost_weight`, whereas the emergency mode uses a fixed value of 1:

$$w_{cost}^{ev} = \begin{cases} \text{ev_charging_cost_weight}, & \text{mode} \in \{\text{soft}, \text{hard}\}, \\ 1, & \text{mode} = \text{emergency}. \end{cases}$$

When $w_{cost}^{ev} > 1$, the agent places greater emphasis on avoiding costly EV charging during periods of high electricity prices.

4.5.3 EV Departure Penalty

The EV departure requirement represents a physical mobility constraint, requiring the EV to reach the target SoC before departure. In the reward function, the departure penalty converts this requirement into a learning signal under the soft mode. The SoC deficit at the departure time step is defined as

$$g_{i,t}^{dep} = \max(0, SOC_{req}^{ev} - SOC_{i,t+1}^{ev}), \quad t_d = t^{dep}.$$

The corresponding penalty is

$$L_{i,t}^{ev,dep} = w_{dep}^{ev} (g_{i,t}^{dep})^2.$$

This penalty is activated only at the departure time step.

4.5.4 EV Progress Penalty

Experimental results indicate that applying the departure penalty only at the departure time step is still insufficient to consistently achieve the required departure SoC. To solve this problem, an additional progress penalty can be enabled in the soft EV experiments. This penalty provides a linear reference trajectory from the arrival SoC to the required departure SoC throughout the EV connection window. Let $\alpha_t \in [0, 1]$ denote the normalized progress within the current connection window. Then,

$$SOC_t^{ev,prog} = SOC_{arr}^{ev} + \alpha_t (SOC_{req}^{ev} - SOC_{arr}^{ev}).$$

if the EV SoC falls below this reference trajectory, a penalty is applied to encourage the agent to gradually approach the departure target during the night connection window. The progress gap can be expressed as

$$g_{i,t}^{prog} = \max(0, SOC_t^{ev,prog} - SOC_{i,t+1}^{ev}),$$

The corresponding penalty is

$$L_{i,t}^{ev,prog} = w_{prog}^{ev} (g_{i,t}^{prog})^2.$$

This term encourages the EV to gradually approach the required departure SoC throughout the overnight connection window, rather than postponing the entire charging demand until the final time steps before departure.

4.5.5 Price-Aware Penalty

During the experiments, it was further observed that increasing the EV costweight may cause competition between the EV reward and the battery reward. Moreover, the EV is still unable to completely avoid charging during periods of high electricity prices. To encourage the EV to avoid expensive charging periods, a price-aware penalty is introduced in this thesis. First, a quantile threshold is computed from the electricity price profile within the current episode:

$$p^{thr} = Q_q(\{p_\tau\}),$$

where q denotes `ev_price_aware_threshold_quantile`, whose used value is 0.70. When the current electricity price exceeds this threshold, the normalized excess electricity price is defined as

$$\delta_t^p = \frac{\max(0, \tilde{p}_t - p^{thr})}{\max(p^{thr}, 10^{-6})}.$$

The price-aware penalty is then defined as

$$L_{i,t}^{ev,price} = w_{price}^{ev} P_{i,t}^{ev} \delta_t^p \Delta t.$$

When the electricity price does not exceed the threshold, this penalty is zero. When the electricity price exceeds the threshold and the EV is charging, this term becomes a positive penalty, encouraging the agent to shift flexible EV charging demand to periods with relatively lower electricity prices.

4.5.6 Battery Reward and Regularization

The battery reward consists of four components: the economic benefit of battery charging and discharging, the boundary action penalty, the SoC regularization term, and the throughput bonus applied during the early stage of training. The instantaneous economic benefit of the battery is given by

$$r_{i,t}^{bat} = [\max(-P_{i,t}^b, 0) - \max(P_{i,t}^b, 0)] \tilde{p}_t \Delta t.$$

When the battery is already close to its SoC limits but continues to move toward the corresponding boundary, an action boundary penalty is imposed:

$$L_{i,t}^{bat,act} = w_{act}^{bat} |P_{i,t}^b| I_{i,t}^{bd}.$$

where $I_{i,t}^{bd} = 1$ indicates that the battery is discharging near a low SoC limit or charging near a high SoC limit; otherwise, $I_{i,t}^{bd} = 0$. Additionally, the SoC regularization term encourages the battery to stay away from the hard limits:

$$L_{i,t}^{bat,soc} = w_{soc}^{bat} [\max(0, SOC_{low}^{soft} - SOC_{i,t}^b)^2 + \max(0, SOC_{i,t}^b - SOC_{high}^{soft})^2].$$

The throughput bonus for battery applied during the early stage of training is given by

$$r_{i,t}^{bat,thr} = w_t^{thr} |P_{i,t}^b| \Delta t.$$

This term gradually decays with training progress, aiming to prevent the actor from learning overly conservative strategies that avoid using the battery entirely in the early stages of training.

4.5.7 Grid Safety Penalty

The grid safety penalty is derived from the results of the pandapower power flow calculation and consists of the three categories introduced previously: voltage violations, line loading violations, and transformer loading violations, as mentioned before. Let ψ_t^V , ψ_t^{line} , and ψ_t^{tr} denote the raw violation measures for voltage, line loading, and transformer loading, respectively. The corresponding penalty terms are

$$L_{i,t}^V = N w_V \psi_t^V \omega_{i,t}^V,$$

$$L_{i,t}^{line} = w_{line} \psi_t^{line},$$

$$L_{i,t}^{tr} = w_{tr} \psi_t^{tr}.$$

where $\omega_{i,t}^V$ denotes the voltage penalty weight based on the degree of voltage violation for each agent bus; line and transformer penalties are currently the same for all agents. This design ensures that agents not only focus on their own economic benefits but also perceive the impact of their actions on the safety of the distribution grid.

4.5.8 Total Reward

The total reward for each agent can be written as

$$\begin{aligned} r_{i,t} = & r_{i,t}^{bat} - C_{i,t}^{ev} - L_{i,t}^{bat,act} - L_{i,t}^{bat,soc} - L_{i,t}^{ev,dep} - L_{i,t}^{ev,proj} \\ & - L_{i,t}^{ev,price} - L_{i,t}^{ev,proj} - L_{i,t}^{ev,eme} + r_{i,t}^{bat,thr} - L_{i,t}^V - L_{i,t}^{line} - L_{i,t}^{tr}. \end{aligned}$$

Here, $L_{i,t}^{ev,proj}$ and $L_{i,t}^{ev,eme}$ denote the diagnostic penalty terms introduced by the hard projection and emergency charging mechanisms, respectively, which will be discussed in the following sections. Overall, the reward function is designed to simultaneously encourage cost-efficient operation, satisfy the EV mobility requirements, promote appropriate battery utilization, and prevent violations of the distribution network safety constraints.

4.6 Safety Projection for Grid Constraints

The reward penalty represents a soft constraint: it allows the agent to take unsafe actions but penalizes such behavior through the reward function. In contrast, the safety projection acts as a hard correction mechanism applied before action execution. It directly modifies the control actions before they are executed by the environment, thereby moving them closer to the grid-safe feasible region. In the projection-based controllers, the safety projection is employed as a complement to the reward penalty mechanism.

$$x_t^{\text{proj}} = [P_{1,t}^b, \dots, P_{N,t}^b, G_{1,t}^{\text{curt}}, \dots, G_{N,t}^{\text{curt}}].$$

Given the current load, PV generation, and EV charging power, the base net load is

$$P_{i,t}^{\text{base}} = L_{i,t} - G_{i,t} + P_{i,t}^{\text{ev}}.$$

The safety projector uses pandapower to compute finite-difference sensitivities of the bus voltages and line loading with respect to the agents' net power injections around the zero-injection operating point. The corresponding linearized relationships can be written as

$$\begin{aligned} V_t &\approx V^{\text{base}} + S^V \Delta P_t, \\ \lambda_t^{\text{line}} &\approx \lambda^{\text{base}} + S^{\text{line}} \Delta P_t. \end{aligned}$$

where ΔP_t is determined by the battery power and PV curtailment. The projection goal is to satisfy the linearized safety constraints with respect to voltage, line, and transformer limits while making minimal changes to the original actions.

Each linearized constraint can be written in half-space form

$$r_j^T x_t^{\text{proj}} \leq b_j.$$

If the current solution x_t^{proj} violates the corresponding constraint, it is projected sequentially onto each constraint hyperplane as

$$x_t^{\text{proj}} \leftarrow x_t^{\text{proj}} - \frac{\max(0, r_j^T x_t^{\text{proj}} - b_j)}{\|r_j\|_2^2} r_j.$$

After the projection step, the battery power is clipped again to the locally feasible SoC range, while the PV curtailment is clipped to

$$0 \leq G_{i,t}^{\text{curt}} \leq G_{i,t}.$$

This procedure is repeated for several iterations. Physically, the projection mitigates voltage and line loading violations by reducing battery charging, increasing battery discharging, or increasing PV curtailment. Since EV charging is primarily driven by the user’s mobility requirements, this thesis treats the EV constraints separately from the grid safety projection.

4.7 EV Departure Requirement Handling

This thesis uses three methods for handling the EV departure requirement: the soft, hard, and emergency charging strategies. All three approaches employ the same physical EV model, but they differ in terms of action execution and reward feedback.

4.7.1 Soft Mode

Soft mode directly executes the EV charging power requested by the actor, and guides policy learning through departure penalties, progress penalties, or costweight penalties in the reward function. The executed power is

$$P_{i,t}^{ev} = P_{i,t}^{ev,rl}$$

and is subject to constraints from EV connection, charger power, and SoC headroom.

4.7.2 Hard Mode

In the hard mode, the execution layer checks whether the current EV charging action is sufficient to preserve the feasibility of meeting the required departure SoC. Let K_t^{future} denote the number of remaining time steps during which the EV will remain connected before departure. The maximum achievable increase in SoC over the remaining connection window is

$$\Delta SOC_{i,t}^{future,max} = \frac{K_t^{future} P_i^{ev,max} \Delta t \eta_{ev}}{C_i^{ev}}.$$

If the EV is not charged sufficiently at the current time step, it may become impossible to reach SOC_{req}^{ev} before departure, even if the charger subsequently operates at its maximum power. Therefore, the hard mode computes the minimum required charging power at the current time step as

$$P_{i,t}^{ev,req} = \max \left(0, \frac{(SOC_{req}^{ev} - \Delta SOC_{i,t}^{future,max} - SOC_{i,t}^{ev}) C_i^{ev}}{\Delta t \eta_{ev}} \right).$$

The finally executed charging power is then determined by

$$P_{i,t}^{ev} = \min \left(P_i^{ev,max}, \max(P_{i,t}^{ev,rl}, P_{i,t}^{ev,req}) \right).$$

The physical meaning is that the control policy is free to select a charging power higher than the minimum required value, but it is not allowed to choose a charging power below the minimum necessary level.

4.7.3 Emergency Charging

Emergency charging is designed to compensate for the potential SoC deficiency that may occur under the EV soft mode. It is activated only within an emergency window immediately before the EV departure time. Let the emergency window consist of K^{eme} time steps. When the EV enters the emergency window and its SoC remains below the required target, the system computes the required charging power based on the remaining SoC deficit:

$$E_{i,t}^{need} = \max(0, SOC_{req}^{ev} - SOC_{i,t}^{ev})C_i^{ev},$$

$$P_{i,t}^{ev,eme} = \min \left(P_i^{ev,max}, \frac{E_{i,t}^{need}}{K_t^{rem} \Delta t \eta_{ev}} \right).$$

The finally executed power is then determined by

$$P_{i,t}^{ev} = \max(P_{i,t}^{ev,rl}, P_{i,t}^{ev,eme}).$$

This mode performs emergency charging only when the departure time is approaching. Therefore, the emergency mode is simpler to implement, but it may result in a large concentration of charging power during the final stage before departure.

4.8 MADRL Training Algorithm

This thesis adopts a multi-agent actor-critic framework based on the centralized training and decentralized execution paradigm. Each agent is equipped with an independent actor:

$$a_{i,t} = \pi_i(o_{i,t}; \theta_i),$$

where $o_{i,t}$ denotes the actor observation of agent i . During training, a centralized critic is employed, which receives the observations and the actions of all agents:

$$Q_i(o_{1:N,t}, a_{1:N,t}; \phi_i).$$

To reduce Q-value overestimation, the code employs twin critics for each agent:

$$Q_{i,1}, \quad Q_{i,2}.$$

Replay buffer saves transition:

$$(o_t, a_t, R_t^{(n)}, o_{t+n}, d_t),$$

where $R_t^{(n)}$ is the n-step return:

$$R_{i,t}^{(n)} = \sum_{k=0}^{n-1} \gamma^k r_{i,t+k}.$$

Here, $n = 48$ corresponds to an accumulated reward over approximately 12 hours. The critic target is given by

$$y_{i,t} = R_{i,t}^{(n)} + \gamma^n (1 - d_t) \min(Q'_{i,1}(o_{t+n}, a'_{t+n}), Q'_{i,2}(o_{t+n}, a'_{t+n})).$$

Target actions are added with TD3-style target policy smoothing noise:

$$a'_{t+n} = \text{guard}(\pi'(o_{t+n}) + \epsilon).$$

where **guard** denotes the local SoC feasibility check and optional safety projection.

Critic loss is

$$L_i^Q = (Q_{i,1}(o_t, a_t) - y_{i,t})^2 + (Q_{i,2}(o_t, a_t) - y_{i,t})^2.$$

During the actor update, only the i -th action in the joint action vector is replaced:

$$a_{i,t}^{pol} = \text{guard}(\pi_i(o_{i,t})),$$

$$L_i^\pi = -Q_{i,1}(o_t, a_{1,t}, \dots, a_{i,t}^{pol}, \dots, a_{N,t}).$$

Target network uses soft update:

$$\theta'_i \leftarrow (1 - \tau_{\text{soft}})\theta'_i + \tau_{\text{soft}}\theta_i,$$

$$\phi'_i \leftarrow (1 - \tau_{\text{soft}})\phi'_i + \tau_{\text{soft}}\phi_i.$$

During training, both exploration noise and feasible random exploration are employed. The former adds random perturbations to the actor output, while the latter samples locally feasible battery actions with a certain probability during the early stage of training, thereby helping the agent explore the feasible battery SoC operating region.

4.8.1 Training Pseudocode

The overall MADRL training procedure adopted in this thesis can be summarized by the following pseudocode:

1. Initialize N actors, N twin critics, the corresponding target networks, and the replay buffer.
2. Reset the parallel training environments and obtain the initial observations.
3. For each environment step:
 - a) Each actor generates a raw action based on its current observation.
 - b) Add exploration noise and perform the local feasibility check.
 - c) If a projection-based controller variant is enabled, perform the safety projection.
 - d) Execute the actions in the environment, update the battery SoC, EV SoC, and net load, and perform the power flow calculation using pandapower.
 - e) The environment returns the reward, the next observation, and the done flag.
 - f) Store the transition in the replay buffer and compute the n -step return.
 - g) Once the replay buffer contains sufficient samples, draw a mini-batch to update the critics.
 - h) Update the actors and target networks every predefined number of training steps.
4. Repeat the above procedure until the specified number of training episodes is reached.

4.9 Baseline Controllers

To evaluate the performance of the proposed MADRL approach, this thesis also implements optimization-based baseline controllers, including Local MPC and ADMM-MPC.

4.9.1 Local MPC

Local MPC solves an independent receding-horizon optimization problem for each prosumer. It optimizes its own battery operation, PV curtailment, and EV charging schedule over the prediction horizon. For prosumer i , the primary decision variables of the Local MPC formulation include

$$c_{i,\tau}, d_{i,\tau}, G_{i,\tau}^{curt}, P_{i,\tau}^{ev}, E_{i,\tau}^b, E_{i,\tau}^{ev}.$$

where c and d denote the battery charging power and discharging power, respectively, with

$$P_{i,\tau}^b = c_{i,\tau} - d_{i,\tau}.$$

Function for the battery is:

$$E_{i,\tau+1}^b = E_{i,\tau}^b + \eta_b c_{i,\tau} \Delta t - \frac{d_{i,\tau} \Delta t}{\eta_b}.$$

Function for the EV is:

$$E_{i,\tau+1}^{ev} = E_{i,\tau}^{ev} + \eta_{ev} P_{i,\tau}^{ev} \Delta t.$$

Local net load is:

$$P_{i,\tau}^{net} = L_{i,\tau} - G_{i,\tau} + G_{i,\tau}^{curt} + c_{i,\tau} - d_{i,\tau} + P_{i,\tau}^{ev}.$$

Local MPC objective function mainly minimizes the battery charging cost, discharging benefit, and EV charging cost:

$$\min \sum_{\tau} \tilde{p}_{\tau} \Delta t (c_{i,\tau} - d_{i,\tau} + P_{i,\tau}^{ev}) + \epsilon_b \Delta t (c_{i,\tau} + d_{i,\tau}) + w_{dep}^{ev} g_i^2.$$

where ϵ_b is a small throughput tie-breaker, used to avoid unnecessary simultaneous charging and discharging or equivalent solutions; g_i is the EV departure gap. The advantage of Local MPC is that it has a small problem size and fast solution time; its disadvantage is the lack of grid coordination across agents.

4.9.2 ADMM-MPC

ADMM-MPC introduces distributed coordination based on Local MPC. It tends to constrain the impact of aggregated net load on transformers. Each agent still solves its local subproblem, but additionally introduces local net load contribution

$$q_{i,\tau} = \alpha_{i,\tau} P_{i,\tau}^{net}.$$

Here, $\alpha_{i,\tau} = 1$. The global coordination objective is to ensure that all agents' contributions satisfy the transformer aggregation bounds:

$$-\bar{P}^{tr} \leq \sum_i q_{i,\tau} \leq \bar{P}^{tr}.$$

ADMM adds a quadratic penalty term to the local objective function :

$$J_i^{ADMM} = J_i^{local} + \frac{\rho}{2} \|\alpha_i P_i^{net} - z_i + u_i\|_2^2.$$

where z_i is the coordination variable, u_i is the scaled dual variable, and ρ is the ADMM penalty parameter. The solution process iterates between local optimization and aggregation projection:

$$\begin{aligned} z &\leftarrow \Pi_{[-\bar{P}^{tr}, \bar{P}^{tr}]}(q + u), \\ u &\leftarrow u + q - z. \end{aligned}$$

Compared with Local MPC, ADMM-MPC provides a certain degree of system-level coordination.

4.9.3 Rule-Based Controller

In addition to the two optimization-based baseline approaches described above, this thesis also implements a rule-based EV and battery controller as a simple reference method. This controller provides a baseline strategy that does not rely on either learning algorithms or optimization solvers. The implemented controller is denoted as `rule_based_ev_with_battery`.

The EV charging rule depends only on the EV connection status and whether the EV SoC has reached its upper limit. If the EV is not connected, no charging is performed. If the EV is connected and $SOC_{i,t}^{ev} < SOC_{\max}^{ev}$, the EV is charged at the maximum charging power:

$$P_{i,t}^{ev} = \begin{cases} P_i^{ev,\max}, & A_t^{ev} = 1 \text{ and } SOC_{i,t}^{ev} < SOC_{\max}^{ev}, \\ 0, & \text{otherwise.} \end{cases}$$

This EV rule represents a simple charging behavior: once the vehicle arrives home, it immediately charges at the maximum power of the residential charger until the EV SoC reaches the upper limit. In addition, the stationary battery is controlled by a price-threshold rule based on the current electricity import price and the quantiles of the training electricity price distribution. Let $\tilde{p}_t$ denote the current electricity import price, and let p^{low} and p^{high} denote the lower and upper quantile thresholds of the electricity price series used during training:

$$p^{low} = Q_{q_{low}}(\tilde{p}^{train}),$$

$$p^{high} = Q_{q_{high}}(\tilde{p}^{train}).$$

Based on these two thresholds, the battery power is determined as

$$P_{i,t}^b = \begin{cases} \alpha_c P_i^{b,max}, & \tilde{p}_t \leq p^{low}, \\ -\alpha_d P_i^{b,max}, & \tilde{p}_t \geq p^{high}, \\ 0, & p^{low} < \tilde{p}_t < p^{high}. \end{cases}$$

where $\alpha_c \in [0, 1]$ and $\alpha_d \in [0, 1]$ denote the charging and discharging ratios, respectively. In the current implementation, both parameters are set to 1. Consequently, the battery is commanded to charge at its maximum power during low-price periods, discharge at its maximum power during high-price periods, and remain idle when the electricity price falls between the two thresholds. The resulting battery action is still processed by the battery SoC feasibility mapping in the environment. Therefore, even if the rule requests maximum charging or discharging power, the executed battery power will never violate the SoC limits $[SOC_{min}^b, SOC_{max}^b]$.

The physical meaning of this controller is: electrical energy is stored during periods of low electricity prices and released during periods of high electricity prices, while EV charging follows a fixed immediate-charging rule. The controller requires neither future forecasts nor the solution of an optimization problem, resulting in a very low computational cost. Compared with the MADRL and MPC approaches, it serves as a baseline that requires neither learning nor optimization.

5 Experimental Setup

This chapter describes the data, device parameters, training parameters, grid configuration, and evaluation metrics used in the experiments in this thesis.

5.1 Dataset and Simulation Period

The data used in this thesis are located in the project directory `datasets/prosumer/`. In accordance with the general load model in Section 3.2, the load-component set used in the experiments is

$$\mathcal{C}^L = \{\text{household}, \text{heat pump}\}.$$

The corresponding data are read from `household.csv` and `heatpump.csv`. The selected single-family-house profiles are SFH12, SFH18, and SFH20 for the three prosumers, respectively. Each component is multiplied by the same prosumer-specific load scaling factor,

$$s_1^L = s_2^L = s_3^L = 20.$$

Timestamps are read as Coordinated Universal Time and converted to the **Europe/Berlin** time zone so that the load, PV, electricity-price, and EV connection schedules use a common local-time reference.

Training and evaluation use different years and date ranges. The current configuration is shown in Table 5.1.

During the evaluation phase, 15 evaluation episodes are selected by default. Each episode corresponds to the operation process of one day, where the controller reads observations and outputs actions at each time step, and the environment updates the system state.

5.2 Battery and EV Parameters

Battery parameters are shown in Table 5.2.

Table 5.1: Dataset and Simulation Time Period.

Item	Setting	Description
Data Directory	datasets/prosumer	Load, PV, and electricity price data
Load Components	Household, heat pump	Summed for each prosumer
Load Scale s_i^L	20, 20, 20	Applied to both load components
Time Zone	Europe/Berlin	Common local-time reference
Training Year	2019	Used for LSTM and MADRL training
Training Dates	2019-01-01 to 2019-12-31	Main training data period
Evaluation Year	2020	Used for controller evaluation
Evaluation Dates	2020-04-01 to 2020-04-15	Main evaluation data period
Time Step	0.25 h	15-minute resolution
Daily Steps	96	One natural day episode
Load Profiles	SFH12, SFH18, SFH20	Three prosumers

Table 5.2: Stationary battery parameters.

Parameter	Value	Description
C_i^b	100 kWh	Battery capacity for each prosumer
$SOC_{\min}^b$	0.05	Minimum battery SoC
$SOC_{\max}^b$	0.95	Maximum battery SoC
SOC_0^b	0.05	Initial battery SoC for evaluation
η_b	0.95	Battery charging/discharging efficiency
$r_{\max}^b$	0.5	Maximum charging/discharging rate
$p_i^{b,\max}$	50 kW	Maximum charging/discharging power

The maximum power of the battery is given by its capacity and maximum charging rate:

$$P_i^{b,\max} = r_{\max}^b C_i^b = 0.5 \times 100 = 50 \text{ kW}.$$

This setting allows the battery to charge when prices are low or PV generation is sufficient, and discharge when prices are high or load is significant, while keeping the SoC within the bounds $[0.05, 0.95]$.

EV parameters are shown in Table 5.3.

Table 5.3: EV parameters.

Parameter	Value	Description
C_i^{ev}	60 kWh	Battery capacity for each EV
$SOC_{\min}^{ev}$	0.10	Minimum EV SoC
$SOC_{\max}^{ev}$	0.95	Maximum EV SoC
SOC_{arr}^{ev}	0.30	SoC at home arrival
SOC_{req}^{ev}	0.90	Required SoC at departure
$P_i^{ev,\max}$	11 kW	Maximum charging power for home charging
η_{ev}	0.95	EV charging efficiency
t^{arr}	72	Home arrival time (18:00)
t^{dep}	28	Home departure time (07:00)

5.3 Training Parameters

5.3.1 LSTM Training Parameters

The LSTM forecasting module predicts the electricity price, load demand, and PV generation. The historical input window is set to

$$h = 192,$$

corresponding to the previous 48 hours, while the prediction horizon is

$$H = 49,$$

corresponding to the current time step and approximately the following 12 hours. The LSTM training parameters are summarized in Table 5.4.

Table 5.4: LSTM forecasting model training parameters.

Parameter	Value
history window	192
sequence length	49
batch size	1024
train ratio	0.70
validation ratio	0.15
price epochs	50
price hidden size	128
price layers	2
load epochs	20
load hidden size	64
PV epochs	20
PV hidden size	64

5.3.2 MADRL Training Parameters

The training parameters of the MADRL framework are summarized in Table 5.5.

The MADRL approaches are referred to by readable method names throughout the thesis. “Base” denotes a controller without an additional reward-shaping or grid-projection mechanism, while “Projected” indicates that grid safety projection is applied before action execution. Table 5.6 provides the one-to-one correspondence between the thesis terminology and the labels stored in the experimental results and plotting notebooks.

5.4 Network Configuration

The network security limits adopted in the experiments are summarized in Table 5.7.

5.5 Evaluation Metrics

Evaluation metrics can be divided into four categories: economic metrics, grid safety metrics, EV departure SoC metrics, and diagnostic metrics for the relationship between EV/battery and electricity prices. These metrics are used to evaluate the controller’s performance from different perspectives, but this chapter only defines the metrics without

Table 5.5: MADRL training parameters.

Parameter	Value
action dimension	3
hidden dimension	256
discount factor γ	0.999
soft update τ_{soft}	0.01
policy update frequency	2
policy noise	0.2
noise clip	0.5
train episodes	1000
number of environments	1
replay buffer size	50000
batch size	128
actor learning rate	10^{-4}
critic learning rate	10^{-4}
n-step return	48
exploration noise	0.4 to 0.2

Table 5.6: Correspondence between readable MADRL method names and experimental result labels.

Thesis name	Experimental result label
Base Soft	Base_soft
Cost-Weighted Soft	Base_Costweight_soft
Progress Soft	Base_Progress_soft
Projected Progress Soft	Projection_Progress_soft
Emergency Projected Soft	Projection_EV_EME_soft
Emergency Soft	Base_EME_soft
Base Hard	Base_hard
Cost-Weighted Projected Hard	Projection_Costweight_hard
Price-Aware Projected Hard	Projection_Priceaware_hard
Projected Hard	Projection_hard
Projected Hard (EV SoC Upper Limit 0.91)	Projection_hard (SoC0.91)

Table 5.7: Network configuration and security limits.

Parameter	Value
SimBench code	1-LV-rural1--0-sw
agent bus ids	12, 4, 2
power flow solver	Newton–Raphson
$V_{\min}$	0.95 p.u.
$V_{\max}$	1.05 p.u.
line loading limit	100%

discussing specific results.

5.5.1 Economic Metrics

The economic metrics are used to evaluate the operating cost of the controller and the economic benefits achieved through energy storage. The main metrics include:

$$\begin{aligned}\mathcal{C}_{total}^{bat} &= \sum_{t,i} \max(P_{i,t}^b, 0) \tilde{p}_t \Delta t - \sum_{t,i} \max(-P_{i,t}^b, 0) \tilde{p}_t \Delta t, \\ \mathcal{C}_{total}^{ev} &= \sum_{t,i} P_{i,t}^{ev} \tilde{p}_t \Delta t, \\ \mathcal{C}^{total} &= \mathcal{C}_{total}^{bat} + \mathcal{C}_{total}^{ev} + \mathcal{C}_{total}^{pv}.\end{aligned}$$

Besides the total cost, the records also include storage charge cost, storage discharge revenue, storage profit, and episode reward.

5.5.2 Grid Safety Metrics

Grid safety metrics are derived from the pandapower power flow results. They mainly include:

$$\begin{aligned}&\min V_{b,t}, \quad \max V_{b,t}, \\ &\text{voltage violation steps,} \\ &\text{feeder netload ramp mean abs.}\end{aligned}$$

where voltage violation steps represents the number of time steps where at least one bus voltage violates the bounds; feeder netload ramp is used to measure the severity of changes in the feeder’s net load.

5.5.3 EV Departure SoC Metrics

EV metrics are used to evaluate whether users' travel demands are met. The core metrics include departure SoC, departure gap, EV session energy, and EV session cost. The departure SoC gap is defined as

$$g_i^{dep} = \max(0, SOC_{req}^{ev} - SOC_{i,t^{dep}}^{ev}).$$

If $g_i^{dep} = 0$, it indicates that agent i meets the target SoC when departing; if $g_i^{dep} > 0$, it indicates that the departure demand is not fully satisfied. The evaluation also records the total charging amount for each EV session:

$$E_i^{ev,session} = \sum_{t \in \mathcal{T}^{ev}} P_{i,t}^{ev} \Delta t,$$

5.5.4 Relationship between EV, Battery Power and Electricity Prices

To analyze whether the controller has learned price-aware scheduling, this paper also records the time-aligned sequences of EV charging power, battery charging/discharging power, and import price.

$$P_{i,t}^{ev} \text{ vs. } \tilde{p}_t,$$

$$P_{i,t}^b \text{ vs. } \tilde{p}_t.$$

A controller with good price responsiveness should schedule EV charging and battery charging predominantly during periods of low electricity prices, while battery discharging should occur primarily during periods of high electricity prices.

6 Results and Discussion

This chapter evaluates different kinds of MADRL controllers and analyses their economic performance, EV feasibility, and grid safety. The analysis is organised mainly around four research questions:

- RQ1: Under the same realised demand, PV generation, price trajectories, and physical network model, can MADRL improve upon a simple rule-based controller?
- RQ2: How do a soft departure penalty, a progress-based penalty, and hard feasibility correction trade economic performance against EV departure feasibility?
- RQ3: How does grid-safety projection affect voltage safety, EV feasibility, and battery arbitrage?
- RQ4: Under aligned initial conditions, forecasts, evaluation interval, and reported economic components, what explains the performance gap between MADRL, Local MPC, and ADMM-MPC?

Sections 6.1 and 6.2 first establish the forecasting and training basis. Sections 6.3–6.5 then answer the research questions through two ablations and a benchmark comparison. Section 6.6 consolidates limitations and failure cases that were previously repeated across several sections.

6.1 Forecasting Performance

The LSTM model provides price, household-load, and PV forecasts used by the predictive controllers. Table 6.1 reports the results used by the following experiments. Each series contains 1,440 test samples.

Let y_t denote the ground-truth value of a forecast signal at time step t , $\hat{y}_t$ its LSTM prediction, and N_{eval} the number of evaluation samples. MAE and RMSE are calculated as

$$\text{MAE} = \frac{1}{N_{\text{eval}}} \sum_{t=1}^{N_{\text{eval}}} |\hat{y}_t - y_t|,$$

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{eval}}} \sum_{t=1}^{N_{\text{eval}}} (\hat{y}_t - y_t)^2}.$$

MAE represents the average magnitude of the forecast error, while RMSE assigns greater weight to large deviations and is therefore more sensitive to errors around local peaks and valleys. From the table, the price model achieves an MAE of 0.0273 EUR/kWh and an RMSE of 0.0368 EUR/kWh. Across the three prosumers, household-load MAE ranges from 9.0144 to 10.4812 kW. The shared PV forecast obtains an MAE of 0.4334 kW and an RMSE of 0.8924 kW.

Table 6.1: LSTM forecasting performance on the held-out test data

Signal	Series	MAE	RMSE
Price	Shared price	0.0273	0.0368
Household load (kW)	SFH12	9.7839	16.1256
Household load (kW)	SFH18	9.0144	14.8399
Household load (kW)	SFH20	10.4812	14.4119
PV generation (kW)	Shared PV	0.4334	0.8924

Figure 6.1 compares LSTM electricity price prediction with the ground-truth values and also shows the prediction error over the evaluation period.

From figure 6.1 the forecast captures the general electricity-price trend, but deviations remain around local peaks and valleys. During high-price periods, the predicted values are generally below the corresponding ground-truth peaks. Some low-price valleys are also overestimated, so the predicted difference between maximum and minimum prices is smaller than in the original series. The LSTM forecast therefore has a smoothing effect on short-term price fluctuations.

This smoothing directly may affect MADRL controllers’ results. When a high price is underestimated, the controller can underestimate the economic benefit of battery discharge or reduced grid import. When a low price is overestimated, it can reduce battery or EV charging during an economically favourable interval. Errors around peaks and valleys may therefore weaken the coordination among the battery, EV, and PV even when the general daily trend is predicted correctly. Because MADRL and MPC use the same LSTM forecast family, these errors do not explain their full performance gap, but they limit the price information available to both methods and must be considered when interpreting the subsequent controller results.

Overall, the LSTM forecast is able to capture the general trend of electricity price. Pre-

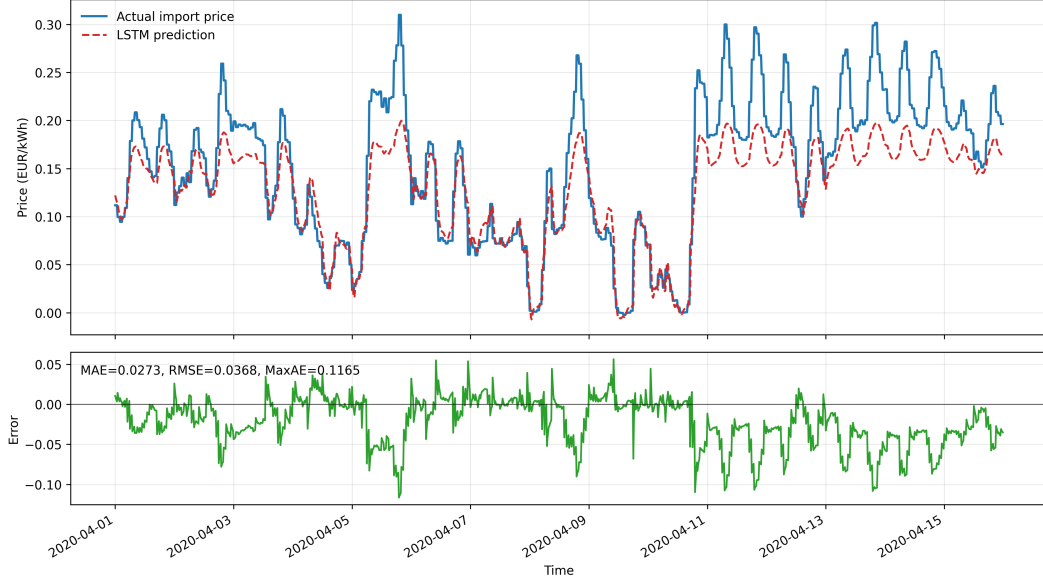


Figure 6.1: LSTM day-ahead price prediction over the evaluation period.

diction errors around local peaks and valleys may however affect the coordination among the energy storage system, EV, and PV generation, thereby influencing the MADRL analysis results.

6.2 MADRL Convergence Analysis

Base Soft was trained for both 100 and 1,000 episodes to investigate the influence of training duration on convergence. Figure 6.2 presents the corresponding training curves. The upper subplot shows the total reward obtained in each episode together with its moving-average curve, while the lower subplot shows the critic loss. To visualise the overall training trend, a moving average over 10 episodes is applied to the 100-episode experiment, whereas a moving average over 50 episodes is used for the 1,000-episode experiment.

As shown in Figure 6.2, when training is limited to 100 episodes, the total reward continues to fluctuate substantially. It improves from -195.19 in the first episode to -50.43 in episode 100. However, the mean reward over the final 10 episodes is still only -61.38, with a standard deviation of 38.90. The large variation at the end of training indicates that the actor and critic have not yet reached stable convergence.

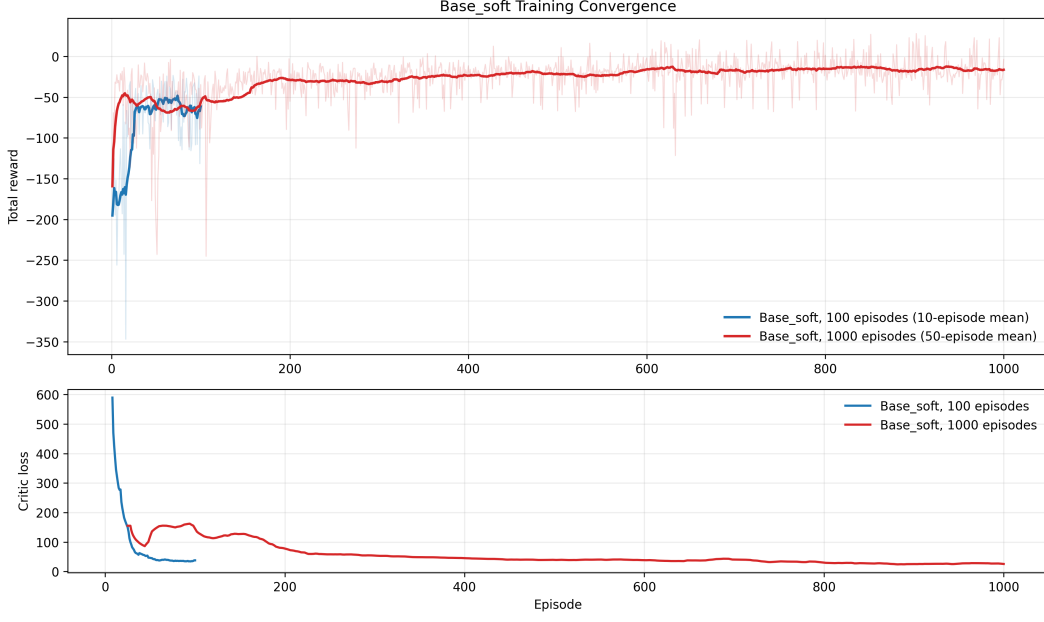


Figure 6.2: Convergence comparison of Base Soft under different numbers of training episodes.

In contrast, the 1,000-episode experiment shows a clearer convergence trend. The total reward improves from -159.30 in the first episode to -3.57 in episode 1,000. Over the final 50 episodes, the mean total reward reaches -16.33, with a standard deviation of 18.96. Compared with the 100-episode experiment, the final mean reward is higher and its variation is smaller. The critic loss also remains more stable during the later training stage. Its final value is 20.03, while the mean critic loss over the final 50 episodes is 25.84.

Battery SoC and EV SoC accumulate over time, while the actions of all prosumers jointly affect the distribution-network state through the power-flow calculation. MADRL therefore requires enough transitions to associate immediate price signals with future SoC states and delayed grid penalties. After only 100 episodes, the critic’s state–action value estimates remain unstable, causing the actor to continue adjusting its preference among battery arbitrage, EV charging, and grid safety. After 1,000 episodes, the more stable critic loss indicates that these temporal and cross-device relationships have been learned more consistently. Consequently, all subsequent MADRL experiments use 1,000 training episodes.

6.3 Ablation of EV-Constraint Handling

This section answers RQ2 by comparing four different MADRL controllers without grid-safety projection, namely: Base Soft, Cost-Weighted Soft, Progress Soft, and Base Hard. Under the soft-constraint approach, battery arbitrage generates economic feedback at every time step and therefore provides a relatively dense reward signal, however, the EV departure penalty is imposed only near the departure time, resulting in a much more sparse reward signal. Here, Base Soft provides the reward-only reference. Cost-Weighted Soft then increases the relative EV-cost weight, Progress Soft adds continuous guidance for EV charging, and Base Hard moves departure feasibility from the reward into action execution.

Table 6.2: Comparison of non-projected Base controllers over 15 evaluation days

Controller	Total cost	EV cost	Storage profit	Mean dep. SoC	Below 0.90	Voltage viol.
Base Soft	57.39	329.78	272.39	0.9460	2	2
Cost-Weighted Soft	251.64	354.39	102.75	0.90	0	0
Progress Soft	188.69	335.68	146.98	0.90	0	1
Base Hard	81.67	287.53	205.87	0.9098	1	1

From the table, Base Soft produces the lowest total cost, 57.39 EUR, and the highest storage profit, 272.39 EUR, but two of the 45 recorded EV departures fall below the 0.90 requirement. Its economic result is explained by reward density. Battery charging and discharging produce a cost signal at every step, so the critic can quickly associate low-price charging and high-price discharging with long-term value. But the departure penalty is activated only near the end of an EV session. The critic must assign that delayed penalty to charging decisions made several hours earlier, making EV feasibility a more difficult credit-assignment problem. The actor therefore learns the dense battery-arbitrage objective more reliably than the sparse EV departure objective. Therefore, Base Soft is economically well-performed, but its low cost is partly obtained by leaving the departure requirement exposed to learning error.

To increase the importance of EV charging demand in the reward function, Cost-Weighted Soft increases the EV-cost weight to 1.5.

This strategy removes 2 departure shortfalls from Base Soft, but the total cost rises by 194.25 EUR and the storage profit falls by 169.64 EUR relative to Base Soft. Despite the larger EV-cost weight, the EV cost increases from 329.78 EUR to 354.39 EUR. Therefore, simply increasing the EV-cost weight can worsen the overall economic performance.

The reason is that adding cost weight does change the magnitude of the EV feedback but cannot tell the actor how to distribute EV charging over the connection window. Because the departure penalty remains delayed, the critic still has difficulty assigning it to individual earlier actions. To avoid a large EV departure SoC penalty, the actor learns a conservative policy and tends to complete EV charging earlier. At the same time, EV charging and battery charging often compete for the same low-price charging window and feeder capacity. As illustrated in Figure 6.3, both devices charge during several highlighted low-price intervals. Cost weighting consequently weakens battery arbitrage without directly improving the timing of EV charging.

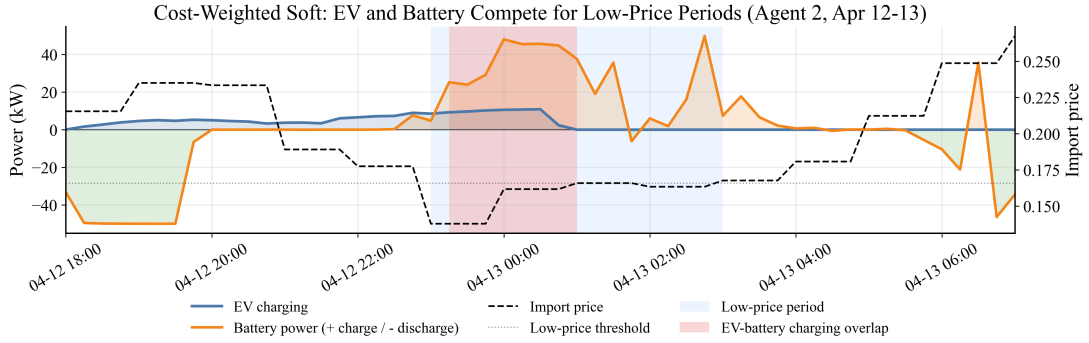


Figure 6.3: EV and battery charging competition during low-price periods under Cost-Weighted Soft. Blue shading denotes low-price periods, while red shading marks intervals in which EV and battery charging overlap.

Therefore, the question is not only about how to reduce costs, but also about how to improve EV departure performance. To investigate the impact of different strategies on EV departure SoC, Figure 6.4 compares the minimum daily departure SoC of the four Base variants. Base Soft reaches a minimum of 0.8782 and crosses below the requirement. Cost-Weighted Soft and Progress Soft remain near 0.95, whereas Base Hard operates close to the 0.90–0.90 feasibility boundary.

Unlike Cost-Weighted Soft, Progress Soft does not simply increase the reward associated with EV charging. Instead, it encourages the EV SoC to approach a time-dependent target progressively throughout the connection window. It also removes both departure shortfalls and reaches a mean departure SoC of 0.95, but it is 62.95 EUR less costly than Cost-Weighted Soft and retains 44.23 EUR more storage profit. At each connected step, the progress term compares the current SoC with the intermediate target and converts the originally sparse departure requirement into dense temporal guidance. The critic can therefore distinguish the long-term value of gradually charging from postponing charging until the final steps. This reduces the credit-assignment difficulty and explains

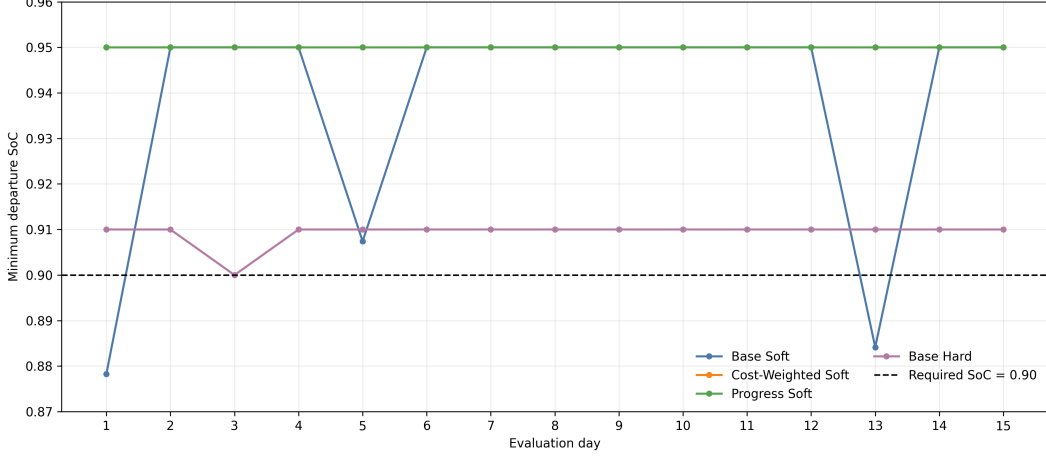


Figure 6.4: Minimum EV departure SoC among the non-projected Base controllers.

why Progress Soft achieves the same departure outcome with a better economic result than Cost-Weighted Soft.

Figure 6.5 illustrates this difference. Cost-Weighted Soft reacts conservatively to the larger penalty scale, whereas Progress Soft distributes charging according to the evolving target. The latter adds temporal information instead of only increasing the consequence of failure.

The same dense guidance also explains the economic loss. EV charging and battery charging compete for the same low-price power and feeder capacity. Progress Soft reserves more of these intervals for EV progress, leaving the battery less freedom to delay charging or discharge at the most profitable times. Its battery charging–discharging price spread decreases from 0.0549 EUR/kWh for Base Soft to 0.0334 EUR/kWh, consistent with the storage-profit reduction. In addition, continuous positive charging guidance tends to move the EV toward the upper SoC region rather than stopping exactly at 0.90. Progress Soft thus improves learned feasibility but may over-satisfy the requirement.

Unlike three soft controllers, Base Hard, which uses hard constraints, turns EV feasibility from a reward-learning problem into an action-execution problem. At each connected time step, the feasibility correction determines the minimum EV charging power required from the remaining energy demand, the remaining connection time, and the charging-power limit. If the actor proposes a lower value, the executed EV power is raised to this minimum. Therefore, the actor is no longer responsible for satisfying the departure requirement. Consequently, Base Hard reduces EV cost to 287.53 EUR and total cost to 81.67 EUR relative to Progress Soft, while preserving 205.87 EUR of storage profit.

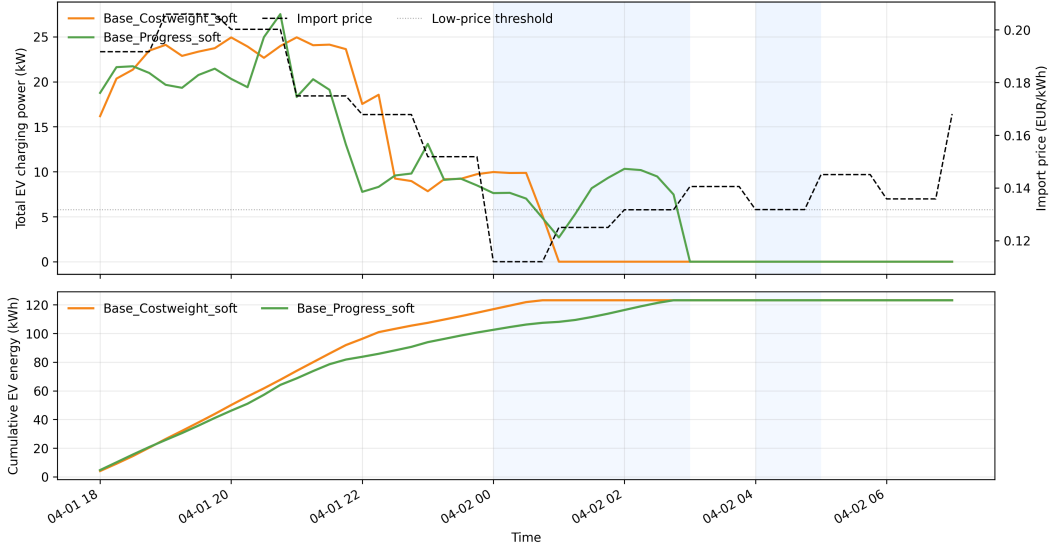


Figure 6.5: EV charging behaviour of Base Soft, Cost-Weighted Soft, and Progress Soft.

Its mean departure SoC of 0.9098 is close to the target because the correction enforces only the lower feasibility boundary rather than continuously rewarding progress toward 0.95.

However, this approach introduces an actor–execution mismatch. The replay transition is generated by the corrected EV power even when the actor proposed another action, so the critic learns the return of an action channel altered by the environment. The archived correction gap of 9.72 kWh confirms that this is not purely theoretical. One recorded departure remains marginally below 0.90, showing that implementation tolerance and the discrete time-step boundary still matter. Base Hard nevertheless provides the best economic–feasibility compromise among the three feasibility-oriented Base variants.

Generally, the four-controller comparison answers RQ2. Base Soft retains the greatest flexibility but can miss departure requirements. Cost weighting changes reward magnitude without adding temporal information. Progress shaping resolves much of the sparse-reward problem but reserves more flexibility for EV charging. Hard correction handles feasibility most directly and performs better economically than both shaped soft variants, although it changes the executed action.

6.4 Ablation of Grid-Safety Handling

This section answers RQ3 by comparing Projected Progress Soft, Cost-Weighted Projected Hard, Price-Aware Projected Hard, and Projected Hard. Every controller uses the same grid-safety projection method, while their EV handling and reward shaping are different from each other. Projected Progress Soft combines projection with dense soft guidance. The other three use hard EV correction. Cost-Weighted Projected Hard scales EV cost, Price-Aware Projected Hard penalises high-price EV charging, and Projected Hard adds neither auxiliary reward term.

Table 6.3: Comparison of projected MADRL controllers over 15 evaluation days

Controller	Total cost	EV cost	Storage profit	Mean dep. SoC	Below 0.90
Projected Progress Soft	462.56	353.55	-109.01	0.9463	1
Cost-Weighted Projected Hard	356.28	289.48	-66.81	0.9084	6
Price-Aware Projected Hard	238.89	291.35	52.46	0.90	3
Projected Hard	186.04	373.42	187.39	0.90	0

All four projected controllers record zero voltage violations. Their minimum/maximum voltages are 0.9746/1.0451 p.u. for Projected Progress Soft, 0.9750/1.0447 p.u. for Cost-Weighted Projected Hard, 0.9722/1.0460 p.u. for Price-Aware Projected Hard, and 0.9707/1.0456 p.u. for Projected Hard. The projection evaluates the joint action against the feeder constraints and replaces an unsafe battery, EV, and PV combination before execution. So the projection corrects the evaluated proposals successfully.

Relative to Progress Soft in Section 6.3, Projected Progress Soft removes the remaining voltage violation, but total cost increases from 188.69 EUR to 462.56 EUR. Relative to Base Hard, Projected Hard removes one voltage violation and the marginal departure shortfall, but total cost increases from 81.67 EUR to 186.04 EUR. These show that network safety improves by the projection, while the actor loses economically favourable actions and the critic must learn from transitions produced after correction.

Here, Projected Progress Soft is the most expensive controller. Its progress signal reserves charging power throughout the EV connection window, while the grid projection can simultaneously modify battery power and PV curtailment. The actor is thus trained toward a dense EV trajectory, but the executed joint action is repeatedly altered by the safety layer. This twofold restriction weakens the association between low-price battery

actions and their realised return. It also leaves one departure below 0.90, demonstrating that soft progress guidance remains a learned preference rather than an execution guarantee.

Cost-Weighted Projected Hard improves EV cost to 289.48 EUR but produces a storage loss of 66.81 EUR and misses six departure records under the strict threshold. This is because that EV-cost weight changes the critic’s reward landscape, the hard correction also changes EV power when the proposal is insufficient, and grid projection subsequently changes the joint action for safety. The actor therefore receives feedback after two execution transformations. A raw action with favourable reward under the learned critic may be mapped to a different feasible point, so stronger EV weighting does not achieve a lower total cost or better recorded feasibility.

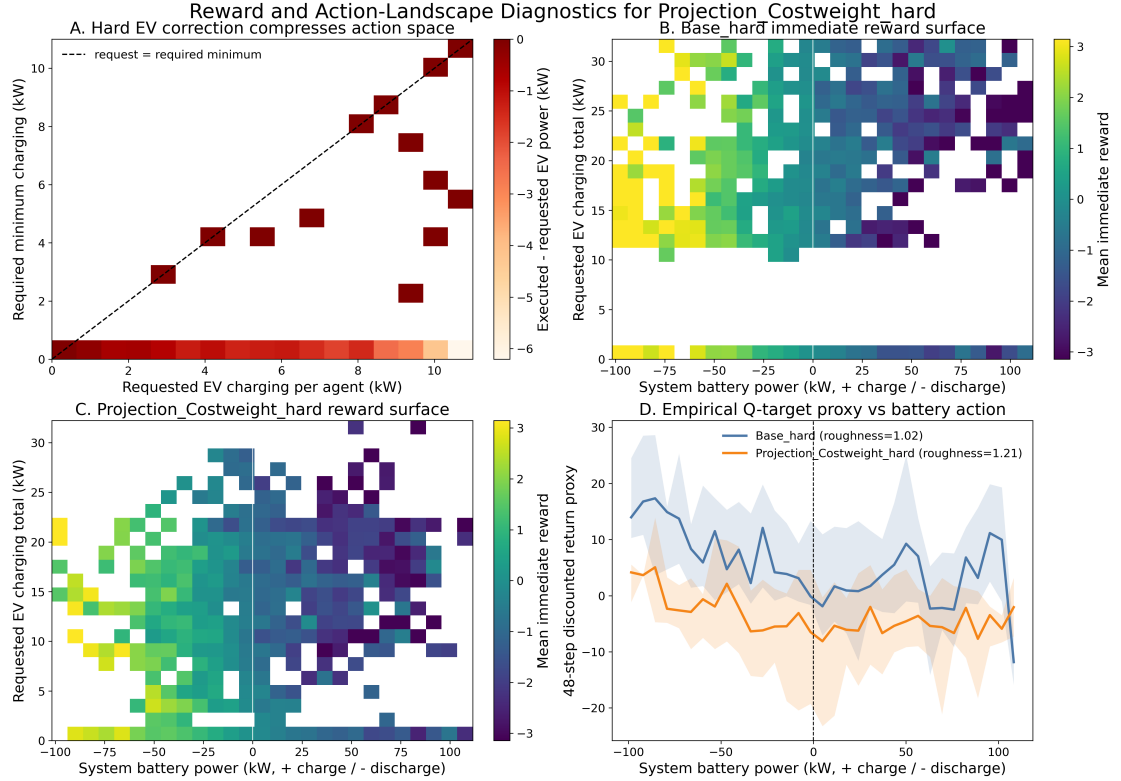


Figure 6.6: Interaction between EV cost weighting, hard correction, and grid projection.

Figure 6.6 illustrates this mismatch. Cost-Weighted Projected Hard simultaneously introduces EV cost reward scaling and grid safety projection. The former changes the relative importance of EV charging in the reward function, while the latter modifies

the battery power and PV curtailment actions to satisfy distribution network safety constraints. Consequently, the effective action space available to the actor becomes narrower than that of Base Hard, and the discrepancy between the raw action and the executed action becomes larger. From the critic’s perspective, the same raw action may be projected into different executed actions under different grid states, resulting in a less smooth Q-value landscape. From the actor’s perspective, actions originally intended for battery arbitrage are more likely to be clipped by the projection. Therefore, although this approach improves distribution network safety, the combination of projection and cost-weight undermines the learning of the battery arbitrage strategy.

Price-Aware Projected Hard reduces total cost to 238.89 EUR and restores a positive storage profit of 52.46 EUR. Its high-price charging share decreases to 38.58%, compared with 39.78% for Cost-Weighted Projected Hard. The improvement is however limited because the price-aware penalty only changes the EV preference. Hard correction still imposes a minimum feasible charge when time is short, and grid projection still constrains the battery action that could otherwise offset EV load. The connection window and remaining energy therefore bound how much charging can be shifted.

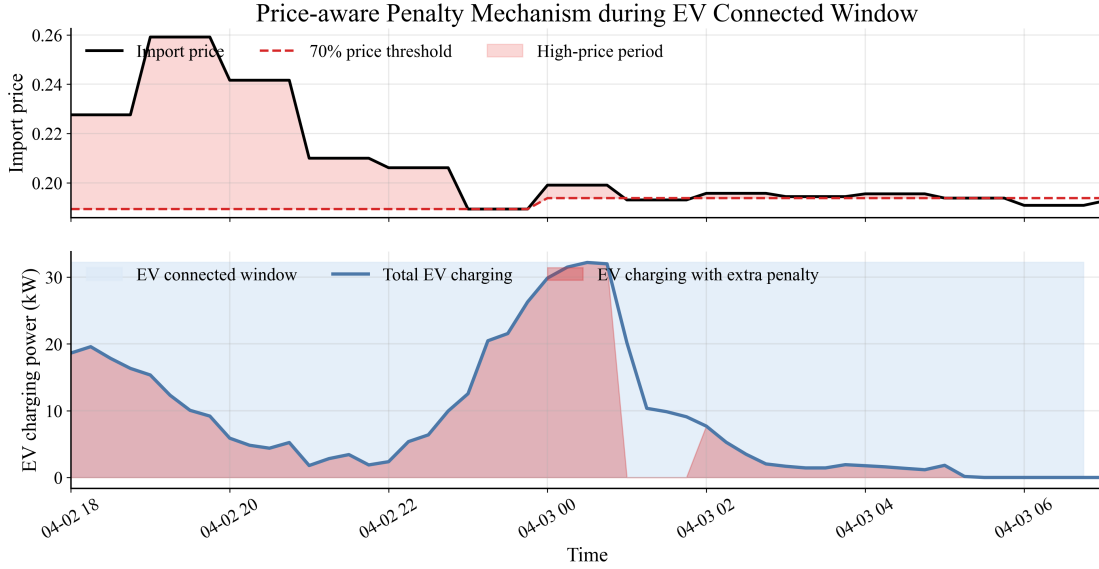


Figure 6.7: Effect of the price-aware penalty under hard EV correction.

As shown in Figure 6.7, the price-aware penalty exhibits a marginal effect. By introducing an additional penalty for EV charging during high-price periods, it encourages the actor to avoid a portion of expensive charging. However, under the hard mode, the minimum EV charging power is already constrained by the departure feasibility

correction, while the connection window is fixed from the evening to the early morning. Consequently, when the remaining scheduling flexibility becomes limited, the price-aware penalty cannot freely shift all charging demand to low-price periods. Meanwhile, the battery actions remain constrained by the grid safety projection and therefore cannot fully compensate for the changes in net load caused by the adjusted EV charging schedule. As a result, although this method reduces the proportion of charging during high-price periods, it does not significantly decrease the EV charging cost nor restore the battery arbitrage profit. In addition, the battery profitability of Price-Aware Projected Hard remains limited. In other words, the price-aware penalty mainly influences the EV charging behavior, whereas the battery actions remain constrained by the safety projection. Although the proportion of EV charging during high-price periods is reduced, the battery fails to recover the arbitrage capability observed under Base Hard. Consequently, the overall economic performance remains unsatisfactory.

Finally, Projected Hard achieves the lowest total cost in this group, 186.04 EUR, and the highest storage profit, 187.39 EUR. It also records zero departures below 0.90. This controller provides a better balance between distribution network safety and battery profitability than the two reward-shaped projection variants. These results indicate that, for the hard constraint controller, directly applying the projection mechanism to enforce the distribution network safety constraints is more effective and stable than additionally introducing the EV cost weight or the EV price-aware penalty.

This indicates that, in the hard controller, less reward shaping can sometimes lead to more stable learning. Projected Hard retains only two execution-layer mechanisms: the EV hard correction ensures mobility feasibility, while the grid safety projection improves distribution network safety. Since no additional EV cost-weight or price-aware penalty is introduced to modify the reward landscape, the actor can more easily preserve its original learning objective of battery arbitrage. However, the EV charging cost of Projected Hard is relatively high. Under this hard projection setting, the EV tends to be charged close to the upper SoC limit. Although this fully satisfies the departure requirement, it also results in unnecessary additional charging energy and a higher proportion of charging during high-price periods. This result also reveals a limitation of the hard correction mechanism. When the feasibility correction is combined with the actor’s own policy, the EV charging behavior may shift from merely satisfying the minimum departure target to approaching the upper SoC limit. In other words, the hard mode guarantees that the departure SoC does not fall below the target value, but it does not inherently prevent overcharging. If the actor continues to request relatively high EV charging power during certain periods while the hard correction ensures feasibility throughout the connection window, the final EV SoC may remain close to the upper limit of 0.95 for an extended period.

The comparison answers RQ3 at two levels. Projection consistently removes the recorded voltage violations, but it narrows the feasible action set and can weaken battery arbitrage. Within that projected action set, additional EV reward shaping does not necessarily improve the final outcome because the actor preference, hard correction, and grid correction can point in different directions. Projected Hard gives the best overall projected result by keeping the reward structure simpler, although its EV overcharging remains a significant limitation.

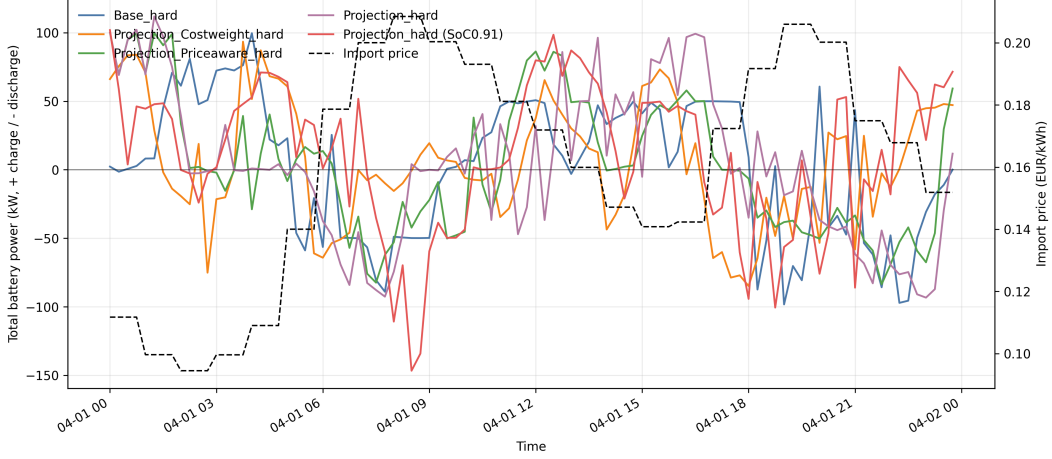


Figure 6.8: Battery power and price for the hard-constraint controllers in the first evaluation episode.

6.5 Benchmark Comparison

Based on Sections 6.3 and 6.4, this section compares four representative MADRL controllers: Base Soft, Progress Soft, Base Hard, and Projected Hard. Base Soft and Base Hard represent the stronger economic results within their respective constraint modes. Progress Soft represents dense EV-feasibility guidance, while Projected Hard represents the combined use of hard EV correction and grid-safety projection. These four controllers are then compared with Local MPC Hard, ADMM-MPC Hard, and rule-based control. The benchmark settings are defined in Chapter 5.

Table 6.4 provides a common reference for comparing the operating characteristics of the selected MADRL controllers and the three baselines.

Among the MADRL controllers, Base Soft represents the economically best perform-

Table 6.4: Economic, EV, and grid comparison of selected MADRL controllers and benchmarks

Controller	Total cost	EV cost	Storage profit	Mean dep. SoC	Below 0.90	Voltage viol.
Base Soft	57.39	329.78	272.39	0.9460	2	2
Progress Soft	188.69	335.68	146.98	0.90	0	1
Base Hard	81.67	287.53	205.87	0.9098	1	1
Projected Hard	186.04	373.42	187.39	0.90	0	0
Local MPC	-36.37	235.68	272.06	0.9000	45	25
ADMM-MPC	-2.86	235.74	238.60	0.9000	45	0
Rule-based	274.01	386.90	112.89	0.90	0	3

mance. Its reward is dominated by dense battery feedback, while the departure objective is delayed and grid safety is not corrected during execution. It therefore retains the greatest scheduling freedom and learns the strongest battery-arbitrage behaviour, but this flexibility is accompanied by EV and voltage violations. Progress Soft shifts the operating point toward EV feasibility by replacing a sparse terminal signal with continuous progress guidance. This improves charging consistency, although EV charging then occupies part of the low-price capacity that would otherwise be available for battery arbitrage. It is therefore the more balanced soft controller, but network feasibility remains a learned outcome rather than an execution guarantee.

Base Hard reaches a different compromise. By embedding the minimum feasible EV charging action in the execution layer, it removes much of the burden of learning departure feasibility from delayed rewards and preserves more economic flexibility than Progress Soft. However, the correction acts only on the EV requirement and does not coordinate the full feeder state. Projected Hard adds this missing network layer: hard correction protects EV reachability, while grid projection corrects unsafe joint actions. It consequently provides the most complete feasibility treatment among the four MADRL controllers, although the additional correction narrows the economically available action space and can produce actor–execution mismatch.

Compared with rule-based control, MADRL has the advantage of adapting its continuous actions to the combined price, load, PV, battery SoC, EV SoC, availability, and network observations. The rule-based controller applies predefined schedules and thresholds and cannot learn how the value of one device action changes with the simultaneous state of the other resources. The selected MADRL controllers therefore demonstrate that learned resource coordination can improve upon fixed local logic. Among them,

Projected Hard adds an explicit safety layer to this adaptive scheduling capability and is the strongest answer to RQ1 when economic performance, mobility, and network operation are considered together.

Comparing with MPC, MPC directly solves a finite-horizon optimisation problem at every control step, using updated forecasts to allocate EV and battery energy while enforcing the constraints included in that problem. However, MADRL learns the same temporal preference indirectly through rewards, critic estimates, exploration, and replay data. Once training is complete, MADRL requires only a policy forward pass and feasibility handling, whereas MPC must solve a new optimisation problem online.

The hard EV correction and grid projection further explain the difference. In MADRL, the actor proposes an action before hard EV correction and grid projection are applied. The critic may therefore learn from a transition generated by an action different from the raw actor output. MPC instead searches its modelled feasible set directly, so its objective and executed decision remain more closely aligned. Local MPC lacks complete feeder-wide coordination and can combine individually economical schedules into an unsafe joint action. ADMM-MPC exchanges coupling information and improves network coordination, but gives up part of the local economic advantage. Thus, the difference between Local MPC and ADMM-MPC reflects the cost of coordination, whereas the difference between MPC and Projected Hard also includes policy approximation and post-policy correction.

The MPC departure SoC values in Table 6.4 are very close to the minimum requirement of 0.90. Due to small numerical differences, they are counted as below the target. However, these results do not represent significant departure constraint violations. They indicate that MPC avoids unnecessary EV charging by operating close to the lower SoC limit. In contrast, Projected Hard maintains a larger SoC margin above the requirement. Therefore, MPC achieves lower charging costs by staying close to the constraint boundary, while Projected Hard uses a more conservative charging strategy to improve departure feasibility.

Figures 6.9 and 6.10, together with Table 6.5, compare the scheduling behaviour of the selected MADRL controllers. Base Soft shows the strongest battery-arbitrage behaviour. Progress Soft uses more charging flexibility to improve EV charging progress. Base Hard maintains better price-aware scheduling because it only corrects the minimum EV charging power required for departure. Projected Hard also responds to electricity prices, but its actions can be modified by both EV correction and grid projection. Therefore, price responsiveness alone cannot determine the best controller. EV charging energy, departure feasibility, and grid safety must also be considered.

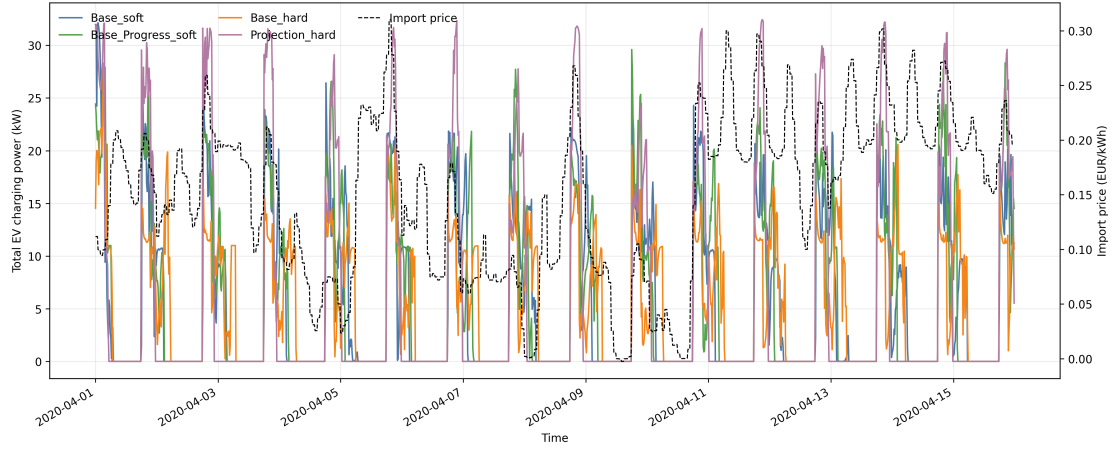


Figure 6.9: EV charging behaviour and electricity price among selected MADRL controllers.

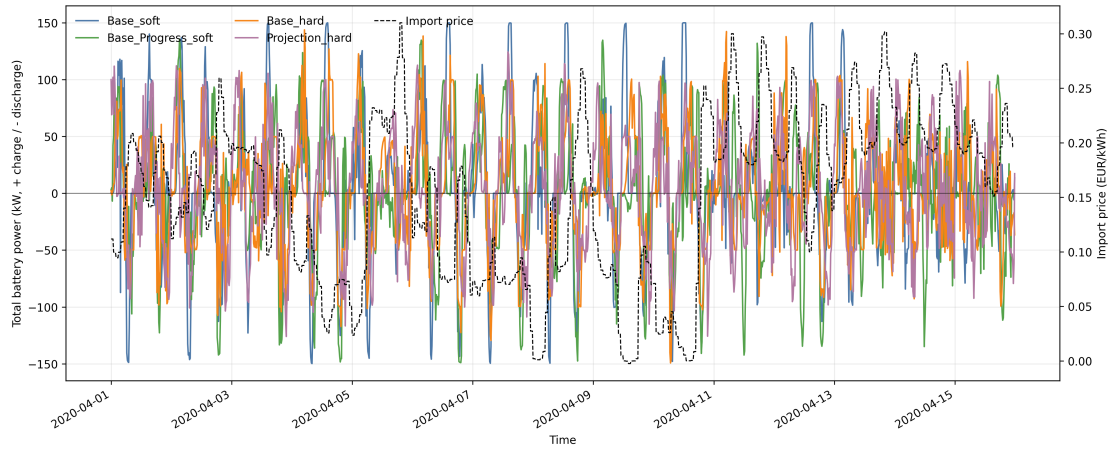


Figure 6.10: Battery operation and electricity price among selected MADRL controllers.

Table 6.5: Price-response indicators among selected MADRL controllers

Controller	EV price	EV energy (kWh)	Low-price share (%)	High-price share (%)	Battery charge	Battery discharge
Base Soft	0.1714	1924.17	27.61	45.99	0.1039	0.1589
Progress Soft	0.1723	1947.67	29.60	45.75	0.1300	0.1634
Base Hard	0.1600	1797.35	33.05	40.34	0.1275	0.1710
Projected Hard	0.1897	1968.33	24.69	56.93	0.1295	0.1742

Overall, Base Soft achieves the lowest economic cost, but its EV and voltage violations make it unsuitable as the final controller. Progress Soft improves EV feasibility through continuous guidance, but it reduces battery-arbitrage performance and still records one voltage violation. Base Hard achieves better economic performance under hard EV constraint handling, but it does not guarantee grid safety. Projected Hard has a higher cost than Base Soft and Base Hard, but it is the only selected MADRL controller that satisfies all 45 departure requirements and records zero voltage violations. MPC achieves better economic performance because it solves an optimisation problem using updated states and forecasts at every control step. However, this requires repeated online optimisation and interaction with the environment. In contrast, MADRL performs the main optimisation during offline training. After training, the actor can directly generate control actions through a forward pass, together with the required feasibility corrections. Therefore, Projected Hard provides faster online decision-making and the best overall trade-off among the selected MADRL controllers when economic performance, EV feasibility, grid safety, and online computational requirements are considered together.

6.6 Limitations and Failure Cases

The results mentioned above establish the main trade-offs, but three limitations restrict the strength and generality of the conclusions.

6.6.1 Emergency Mechanism Failure

Emergency Soft and Emergency Projected Soft were designed to delay EV charging until the remaining connection time became critical. During the 15-day evaluation, Emergency Soft achieves a mean departure SoC of 0.6112 and fails to satisfy all 45

departure requirements, with a total departure gap of 12.9948. Emergency Projected Soft improves the mean departure SoC to 0.8702, but still misses 16 departures and has a total gap of 2.7520.

The main reason is that emergency charging is a late corrective mechanism rather than a continuous training signal. Before it is activated, the critic receives limited guidance on how much charging progress should be completed during the connection window. If the actor delays charging for too long, the remaining energy demand may become greater than the energy that can be delivered within the remaining time. In this case, emergency charging cannot recover departure feasibility even at the maximum charging power. Moreover, the corrective charging actions are concentrated within a short period and may overlap with household demand and battery charging. This increases grid stress, while grid projection may further limit the available EV charging power. Therefore, emergency charging cannot establish a stable long-term charging strategy and is unsuitable as the main soft-constraint mechanism in this thesis.

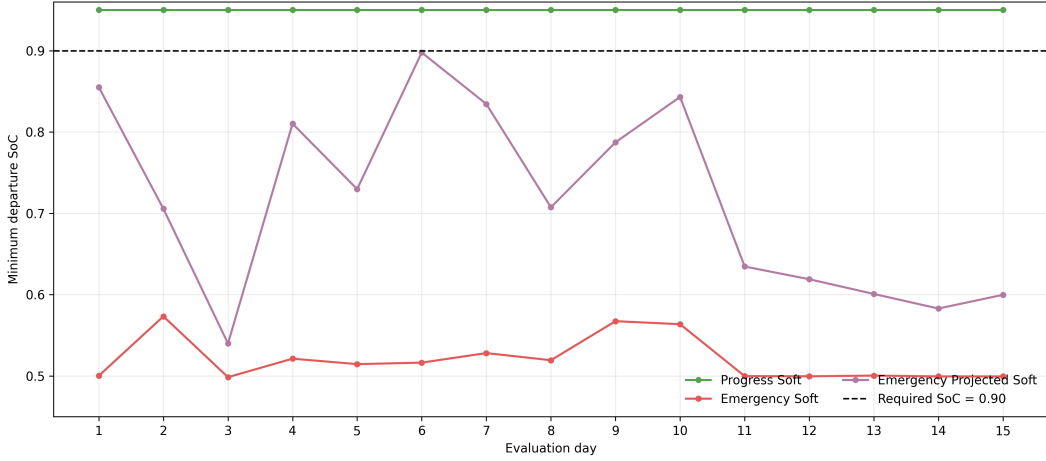


Figure 6.11: Departure-SoC failure of the EME variants relative to the other soft controllers.

6.6.2 Overcharging and Boundary Conventions

From the previous results, Projected Hard tends to charge EVs to the upper SoC region rather than stopping at the 0.90 departure requirement. Hard correction defines the minimum action required to preserve feasibility, but it does not penalise an actor overcharging above this minimum. The grid projection can also retain a conservative charging margin when it corrects the joint EV and battery actions. This behaviour

removes departure shortfalls, but it increases the EV energy and the corresponding charging cost.

To reduce this overcharging phenomenon, an additional Projected Hard variant sets the EV SoC upper limit to 0.90 instead of 0.95. Over the complete evaluation period, this modification reduces the EV charging cost from 373.42 EUR to 288.43 EUR and the EV energy from 1968.33 kWh to 1784.41 kWh. It also reduces the high-price EV charging share from 56.93% to 37.67%. However, the storage profit changes from 187.39 EUR to a loss of 205.82 EUR, causing the total cost to increase from 186.04 EUR to 494.25 EUR. Therefore, reducing EV overcharging does not improve the overall economic performance because the battery arbitrage capability is severely degraded.

Table 6.6: Comparison of Projected Hard with EV SoC Upper Limits of 0.95 and 0.91

<i>(a) Economic and EV-charging indicators</i>						
SoC upper limit	Total cost	EV cost	Storage profit	EV energy (kWh)	EV price	High-price share (%)
0.95	186.04	373.42	187.39	1968.33	0.1897	56.93
0.91	494.25	288.43	-205.82	1784.41	0.1616	37.67
<i>(b) Departure-feasibility and correction indicators</i>						
SoC upper limit	Mean dep. SoC	Below 0.90	Projection gap (kWh)			
0.95	0.9500	0	0.00			
0.91	0.9093	3	13.45			

Table 6.6 separates the intended EV effect from the unintended battery effect. Reducing the upper limit by 0.04 lowers the EV energy by 183.92 kWh, the EV cost by 84.99 EUR, and the weighted EV charging price by 0.0281 EUR/kWh. Nevertheless, storage performance deteriorates by 393.21 EUR, which is substantially larger than the EV-cost saving and explains the 308.21 EUR increase in total cost. The tighter boundary also changes the mean departure SoC from 0.9500 to 0.9093 and introduces a projection gap of 13.45 kWh. The three values counted below 0.90 lie at the numerical feasibility boundary, but they indicate that the reduced upper margin leaves less tolerance for correction and discrete-time effects.

Figures 6.12 and 6.13 compare Agents 0–2 from the evening of 1 April to the morning of 2 April. The EV comparison confirms the intended local effect. In this window, the EV energy of each agent decreases from 41.05 kWh under Projected Hard to 38.53 kWh under the 0.91 upper limit. Projected Hard performs most EV charging before midnight and then stops when the EV is no longer able to absorb further energy. With the 0.91

limit, the charging trajectory is redistributed and terminates at a lower accumulated energy. Hence, the upper limit successfully reduces overcharging; the deterioration does not originate from an increase in EV energy.

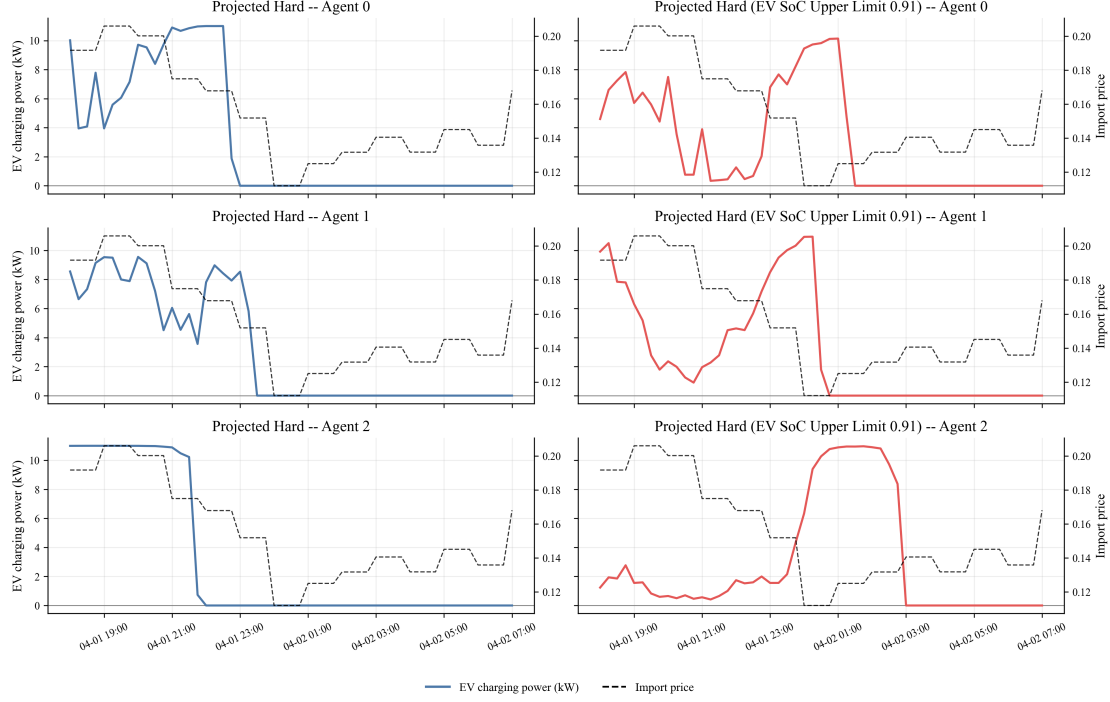


Figure 6.12: EV charging power and import price for Projected Hard and Projected Hard with the EV SoC upper limit of 0.91 during an overnight connection window.

This explains the economic failure. Under Projected Hard, the battery trajectories retain a clearer charging–discharging response to the overnight price variation. After the EV upper limit is reduced, the battery actions become substantially more aggressive and less aligned with the electricity price signal. The battery charging energy increases from 78.64 to 233.70 kWh for Agent 0, from 117.74 to 186.03 kWh for Agent 1, and from 105.63 to 146.12 kWh for Agent 2 in the displayed window. For Agent 0, the discharged energy simultaneously decreases from 143.80 to 105.53 kWh. The figure also shows repeated near-maximum charging actions and charging during relatively high-price periods. Thus, the EV-cost saving is outweighed by a less profitable battery trajectory.

Besides, reducing $\text{SoC}_{\max}^{\text{EV}}$ is not an independent evaluation-time adjustment. Once the EV can absorb less energy, the local net-load trajectory and the complementary re-

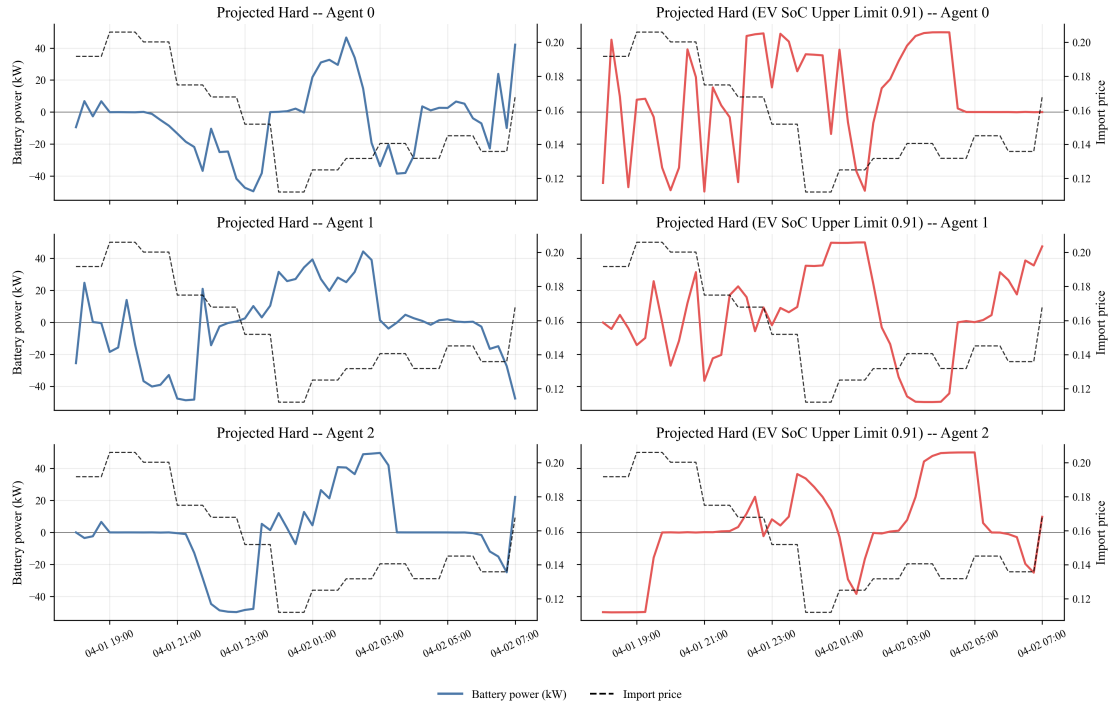


Figure 6.13: Battery charging–discharging power and import price for Projected Hard and Projected Hard with the EV SoC upper limit of 0.91 during the same overnight window. Positive battery power denotes charging.

relationship between EV and battery charging both change. The grid-safety projection consequently modifies the battery actions under a coupling pattern that differs from the one represented during training. Nevertheless, the battery actor still follows a policy learned from the original EV load and SoC boundaries. Its proposed actions are therefore corrected differently, and the resulting battery trajectory can charge at inappropriate prices or fail to discharge sufficiently during high-price periods. The 0.91 experiment consequently demonstrates an actor–projection distribution mismatch rather than a failure of the upper limit to reduce overcharging. A useful implementation of the tighter bound would require retraining, or projection-aware learning with the 0.91 limit included throughout training, so that the battery policy can adapt to the changed net-load and feasible-action distributions.

Overall, the experiments answer the four research questions and show that no controller performs best in every aspect. Progress shaping improves EV feasibility by providing continuous charging guidance. Hard correction satisfies the departure requirement more directly, but it may change the action proposed by the actor. Grid projection prevents the recorded voltage violations, although it reduces scheduling flexibility. Among the selected MADRL controllers, Projected Hard provides the best overall trade-off between economic performance, EV feasibility, and grid safety. It performs better than rule-based control, but its economic performance remains lower than that of the MPC controllers. This difference mainly results from the mismatch between the actors’ proposed action, the corrected action that is finally executed, and the reward used during training.

7 Conclusion and Outlook

7.1 Conclusion

This thesis investigates the building energy management problem of multiple prosumers in a low-voltage distribution network. Each prosumer is equipped with household loads, PV generation, stationary battery energy storage system, and residential EV charging demand. The objective is to coordinate these resources under dynamic electricity prices while considering EV mobility requirements and distribution network safety.

A complete modeling, control, and evaluation framework is established. The system model includes household loads, PV generation, stationary batteries, EV charging, electricity prices, net load calculation, and power flow constraints. The battery and EV models describe the main physical constraints, including SoC dynamics, power limits, EV connection windows, arrival SoC, and departure requirements. Based on this model, the proposed control framework combines LSTM forecasting, MADRL actor-critic control, EV constraint handling, and optional grid safety projection.

The final results show that the LSTM model can capture the main trend of electricity price variations and provides useful forecast information for the controller, and the MADRL training results indicate that a sufficiently long training process is necessary for learning stable scheduling behavior in the presence of battery SoC, EV SoC, electricity price, and grid feedback.

This thesis gives different kinds of controlling methods for EV departure requirements. For EV soft constraints, reward-only control can learn battery arbitrage behavior but can not well satisfy the EV departure SoC requirement. The Progress penalty improves EV feasibility and provides a better balance between EV charging requirements and economic performance than simple cost-weight adjustment. For EV hard constraints, this method can ensure the EV departure requirement at the action execution layer and improve mobility feasibility. However, EV hard handling alone is not sufficient to guarantee grid safety.

Grid safety projection improves distribution network operation by reducing safety violations. At the same time, it may reduce battery arbitrage performance because the

executed actions can differ from the raw actions generated by the MADRL actor. Therefore, the results reveal a clear trade-off between economic performance and grid safety when action projection is applied.

Compared with the rule-based battery controller, MADRL learns more flexible price-aware and resource-coupling policies. Compared with MPC, MADRL still shows weaker economic performance, but it provides fast decentralized execution after training. Overall, the results demonstrate that MADRL is a promising approach for multi-prosumer energy management in low-voltage distribution networks. In the studied scenario, Base Hard is more suitable when economic performance is prioritized, while Projected Hard is more robust when EV feasibility and grid safety are emphasized.

7.2 Outlook

Although this thesis has established a relatively complete framework for multi-prosumer modeling, control, and evaluation, and has systematically compared the performance of MADRL, MPC, and the rule-based battery controller, some directions remain worthy of further investigation.

Projection-aware MADRL training: The experiments in this thesis demonstrate that grid safety projection can significantly improve distribution network safety but may also weaken battery arbitrage. The fundamental reason is that the raw actions generated by the actor are inconsistent with the projected actions ultimately executed by the environment. Future work could incorporate the projection mechanism more directly into the training process. For example, the critic could learn the effects of the projected actions on the reward and the next state, or the actor could directly receive feedback based on the projected actions during training. This would enable the policy to adapt to the ultimately executed actions during learning, thereby reducing the negative impact of action projection on economic performance.

Larger-scale and more heterogeneous distribution network scenarios: This thesis considers a low-voltage distribution network consisting of $N = 3$ prosumers, which is suitable for analyzing the fundamental coupling among EVs, stationary batteries, PV systems, and distribution network safety. However, practical distribution networks may include a larger number of households, more complex feeder topologies, PV systems and batteries with different capacities, diverse EV arrival and departure patterns, and a higher proportion of uncertain loads. Future work could extend the proposed framework to larger SimBench networks or real-world distribution network datasets in order to evaluate the scalability and robustness of MADRL under more complex operating

conditions.

Real-time deployment and hybrid control methods: MPC demonstrates superior economic performance, but its computational burden increases as the number of prosumers and the complexity of constraints grow. In contrast, MADRL offers fast online execution after training but requires extensive simulations and carefully designed reward functions during training. Future work could systematically compare the computation time of MADRL, Local MPC, and ADMM-MPC, and further explore hybrid control approaches. For example, MPC could be used to generate expert trajectories for imitation learning or to warm-start the MADRL policy, thereby combining the optimization capability of MPC with the fast execution of MADRL.

In summary, future work should focus on improving the awareness of MADRL with respect to execution-layer correction mechanisms, enhancing the coordination among EV feasibility, economic performance, and distribution network safety, and validating its generalization capability in larger-scale and more heterogeneous prosumer networks. These extensions will further improve the practical applicability of MADRL-based energy management for residential low-voltage distribution networks.

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